The Limits of Binding in Dual Encoders

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Abstract

Dual-encoder models such as CLIP score an image–caption pair by a single inner product of two independently computed unit vectors, and fail — often near chance — at *binding*: distinguishing "a red car and a blue dog" from "a blue car and a red dog." We give a mathematical account of when this failure is necessary and when it is contingent. Working within the ideal-encoder framework proposed by Kang et al., we first show the relevant axioms are satisfiable, so every impossibility must enter through an added, checkable hypothesis. We then prove three such obstructions. *Depth*: for recursive role-binding codes the swap margin obeys an exact law $m(D) = 2\,b^{-D}$ in the nesting depth D (branching factor b), with a finite-dimension version holding up to one explicitly flagged concentration estimate; crossing the law with the noise floor yields a resolvable depth $D^* = \log N/(2\log b) + O(1)$ — single digits at CLIP scale, the nesting depth of ordinary language. *Objective*: architecture-free throttle theorems showing the contrastive objective's entire reward for binding is confined to a sliver of width $(\pi + \pi_{\text{impl}})\log 2$ plus an exponentially small term, with π the swap-negative rate and π_{impl} the implicit collision rate, and that exactly reversed binding costs only π times the mean binding margin $\mathbb{E}[M]$ — both verified to within 0.006 in simulation. *Geometry*: a smoothness–binding frontier $M(s) \leq 2\sqrt{2\delta}$ in the paraphrase-smoothness gap δ, tight, whose text-only diagnostic we measure across the text encoders of 18 deployed models: every model sits at ≈25–35% of its ceiling, and on 14 of them the induced ceiling tracks SugarCrepe's subset-level difficulty at $r = 0.99$. Binding failure in deployed dual encoders is thus not a dimension or smoothness limit today, but an incentive and code-structure limit — with a proved depth ceiling that remains once those are fixed.

1 Introduction

A *dual encoder* is a pair of neural networks: an image encoder and a text encoder, trained (typically with a contrastive objective) so that matching image–caption pairs receive a high score and mismatched pairs a low one. The score is a single inner product between the two embedding vectors. This architecture — CLIP [1] and its descendants — powers image retrieval, zero-shot classification, and the text conditioning of image generators.

A large empirical literature documents that these models fail at *compositional binding*. Given an image of a red car next to a blue dog, CLIP-class models score the caption "a blue car and a red dog" nearly as high as the correct caption — sometimes higher. Benchmarks built around such *hard negatives* — ARO [2], SugarCrepe [3], Winoground [4] — place strong models near chance (ARO, Winoground) or at their weakest subsets (SugarCrepe's swaps) exactly on the negative types that permute which attribute goes with which object.

Two questions must be separated. *Why do trained models fail today?* — a question about architectures, objectives, and data, answered quantitatively in our companion paper [6]. And: *what limits binding in principle?* — a question about the geometry of the embedding space itself: are there theorems that no amount of scale, data, or training can evade? This paper is about the second question.

The framework we build on is due to Kang et al. [5], who proposed studying *ideal* encoders: instead of asking what a particular network learns, one asks whether *any* assignment of unit embedding vectors could satisfy the inequalities that benchmark competence requires. Their proposal asks the right question; we work with our own formulation of the axioms (Appendix B), stated so that the quantifier structure and the negative families are fully explicit. One structural fact then constrains the whole analysis: *the axioms are satisfiable* (Appendix C) — satisfiable in dimension as low as $N = 2k$ (k the number of atoms in a scene), and realized cheaply at deployed scales by random factored sign codes — so there is no unconditional "dual encoders cannot bind" theorem. Every rigorous account of the failure must be *conditional*: impossibility or limitation *given* something; the scientific content is in how natural and how checkable that something is. Our contributions are three such conditional obstructions, each with its hypothesis stated as a measurable or structural condition, plus the measurements that locate deployed models relative to them.

- **A proved depth ceiling for recursive binding** (§3). For recursive role-binding codes — the classical holographic / vector-symbolic (VSA) composition [10, 11], and the recursive form of pooled codes — the swap margin obeys an *exact* law $m(D) = 2\,b^{-D}$ in nesting depth D, and the replace margin b^{-D}. We prove the law exactly in the zero-crosstalk idealization, and in finite dimension N as a multiplicative-error theorem, modulo one concentration estimate that we state and flag explicitly. The resolvable depth is $D^* = \log N/(2\log b) + O(1)$: at CLIP's $N = 512$ the measured detection floor gives $D^* = 2$ for $b = 2$ (the pure crossing gives 4.5) — the depth of ordinary language. A companion aliasing proposition shows that with self-inverse (shared) roles some deep edits are invisible at *any* dimension; fresh per-level roles repair this, measurably.
- **Throttle theorems for the contrastive objective** (§4). Architecture-free: for *any* scorer, a pairing-blind (bag-of-words) competitor is ε-optimal with $\varepsilon \leq (\pi + \pi_{\mathrm{impl}})\log 2$ plus an exponential tail, where π is the explicit swap-negative rate and π_{impl} the implicit collision rate; and the *exactly reversed* scorer costs precisely $\pi\,\mathbb{E}[M]$ more. At web scale π is vanishing, so the objective's entire reward for binding — and its entire penalty for anti-binding — collapses. Both constants are verified to within 0.006 in simulation.
- **A tight smoothness–binding frontier, measured across deployed models** (§5). If the two swap-related captions embed δ-close to a pairing-neutral anchor (a similarity/paraphrase demand), then the binding margin obeys $M(s) \leq 2\sqrt{2\delta}$, and the constant is exact. The bound induces a text-only diagnostic — frontier *usage*, the ratio of the caption–swap embedding distance d to the ceiling $2\sqrt{2\delta}$ — which we measure on the text encoders of 18 deployed models: every one sits at ≈25–35% of its ceiling, so no deployed model is smoothness-limited; the failure lies in the code and the objective, not the geometry's budget. The per-item ceiling $|M(s)| \leq d(s)$ tracks subset-level difficulty on a real benchmark (SugarCrepe; 14 models, 66,598 decisions) at $r = 0.99$.

We also record what does *not* explain the failure (§6): dimension (the possibility theorem), and exposure-graph connectivity (a natural conjecture we refute — a sign-agnostic matcher generalizes across disconnected exposure components at accuracy 1.0). Full definitions, proofs, protocols, and the negative results are developed self-containedly in the appendices.

2 Setup

Fix finite sets of objects and attributes. An *atom* is a pair (o, a) ("object o has attribute a"); a *k-scene* s is a set of k atoms with distinct objects (and, where swaps are discussed,

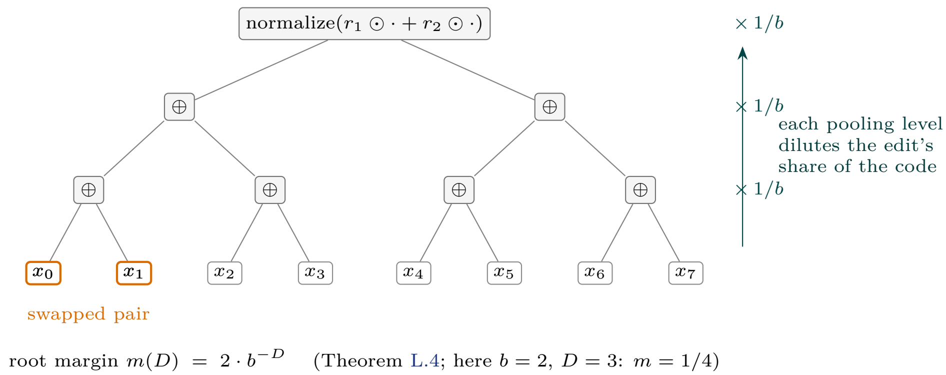


Figure 1: Recursive role-binding (Definition L.1) and the depth law. A deep edit (orange) changes one summand among b at its parent; each further level of normalized pooling multiplies its share of the root code by $1/b$, so the discriminative margin decays geometrically with nesting depth — the recursive continuation of the $\Theta(1/k)$ single-level margin bound of Theorem E.7.

distinct attributes). Each scene has a caption $c(s)$ asserting exactly its atoms and an image I_s depicting them. The *swap* σs exchanges the attributes of two chosen atoms: $c(s)$ and $c(\sigma s)$ contain the *same multiset of words* — we call this shared multiset of object and attribute symbols the scene's *symbol multiset* — and differ only in which attribute is bound to which object: the binding-critical hard negative instantiated by ARO and SugarCrepe (Appendix B maps the benchmark taxonomies onto the formal negative families).

A *dual encoder* is a pair of maps $\mathbf{i}, \mathbf{t}$ into the unit sphere $\mathbb{S}^{N-1}$, scoring a pair by $\langle \mathbf{i}(I), \mathbf{t}(c) \rangle$. Benchmark competence is formalized by two axioms on a *placement* — an arbitrary assignment of unit vectors to images and captions — for a margin $\gamma > 0$ (their full quantifier structure is in Appendix B):

- (C_γ) *Margin retrieval:* for every scene s and every caption c' describing a different scene, $\langle \mathbf{i}(I_s), \mathbf{t}(c(s)) \rangle \geq \langle \mathbf{i}(I_s), \mathbf{t}(c') \rangle + \gamma$;
- (K_k) *Constituent ranking:* the image of a scene scores every *present* object's single-object caption above every *absent* object's.

The quantity every theorem controls is the *binding margin*

$$M(s) \;:=\; \langle \mathbf{i}(I_s),\; \mathbf{t}(c(s)) - \mathbf{t}(c(\sigma s)) \rangle, \tag{1}$$

the swap instance of (C_γ): benchmark accuracy on a swap item is $\mathbf{1}[M(s) > 0]$.

Because sign conventions matter below: all embeddings are unit vectors, $\odot$ is the coordinatewise product, and a *sign vector* has entries ± 1.

3 The depth ceiling of recursive binding

Pooled composition — encoding a set of constituents as the normalized sum of their vectors — has margins $\Theta(1/k)$ in the number k of constituents pooled at one level (Appendix E). Recursion is what makes language productive, and it is where the sharpest law appears. Consider the classical recursive role–filler composition of holographic/vector-symbolic representations [10, 11]: a full b-ary tree of depth D whose leaves carry random unit sign vectors, each internal node encoded as the normalized role-bound sum of its children, $\mathrm{code}(v) = \sum_j r_j \odot \mathrm{code}(\mathrm{child}_j(v)) \,/\, \|\sum_j r_j \odot \mathrm{code}(\mathrm{child}_j(v))\|$, with sign-vector roles r_j either *shared* across levels (the classical convention) or drawn *fresh* per level. The *deep-edit margin* $m(D)$ is $1 - \cos$ between the codes of a tree and a minimal edit of it: a *sibling swap* (two leaves under one parent), a *far swap* (two leaves whose lowest common ancestor is the root), or a *replace* (Appendix L, Definition L.1).

The core of both proofs is one deterministic lemma: a level of role-bound, normalized pooling contracts edit-induced discrepancies by exactly $1/b$, with all error controlled by cross-slot inner products.

Lemma 3.1 (One-level contraction; Appendix L, Lemma L.2)**.** *Let $z_1, \dots, z_b$ and $z'_1, \dots, z'_b$ be unit vectors agreeing outside a set E of* edited slots*; write $g_j := 1 - \langle z_j, z'_j \rangle$ for the* slot deficit *of $j \in E$, and $G := \sum_{j \in E} g_j$. Bind and pool: $P := \sum_j \rho_j \odot z_j$ and $P' := \sum_j \rho_j \odot z'_j$ with sign-vector roles ρ_j, and let the* crosstalk *ε be the largest inner product in magnitude between distinct bound slots. If $(b-1)\varepsilon \le \frac{1}{2}$, then the normalized codes satisfy*

$$\left| \left(1 - \langle P/\|P\|,\, P'/\|P'\| \rangle\right) - \tfrac{G}{b} \right| \;\le\; 8(b-1)\varepsilon.$$

Theorem 3.2 (Exact depth law; Appendix L, Theorem L.4)**.** *In the zero-crosstalk idealization ($\varepsilon = 0$ at every level — the $N \to \infty$ limit of the random model), for either role scheme, provided all bound products appearing in the construction are distinct:*

$$m_{\mathrm{swap}}(D) \;=\; 2 \cdot b^{-D}, \qquad m_{\mathrm{replace}}(D) \;=\; 1 \cdot b^{-D},$$

for both sibling and far swaps. More generally, an edit whose slot deficits at level ℓ_0 sum to G has root margin $G \cdot b^{-(D-\ell_0)}$: the margin depends only on the total deficit and the number of contraction levels, not on where in the tree the edit sits.

The theorem is Lemma 3.1's upward induction; at $\varepsilon = 0$ the contraction is exact, and the finite-N theorem below runs the same induction with a refined form of the lemma whose injected error is proportional to $\sqrt{g}$ (Appendix L, Lemma L.3) — which is what makes the finite-N error multiplicative rather than additive. Two consequences of the mechanism are worth stating: the previously puzzling equality of sibling and far swaps is transparent (a far swap is two replacements in parallel branches, and 1+1 contracted equals 2 contracted); and depth 1 with $b = k$ recovers the single-level constants of Theorem E.7 (swap $2/k$, replace $1/k$).

Theorem 3.3 (Finite dimension; Appendix L, Theorem L.8)**.** *For the fresh role scheme, $D \ge 2$, and $N \ge C\, b^2 D^2 \log(bQ/\eta)$ (where $Q \le b^D + 1$ counts the distinct leaf vectors, the* primitives*), with probability at least $1 - \eta$, simultaneously for every edit above,*

$$m(D) \;=\; m_{\mathrm{ideal}}(D) \cdot (1 \pm \kappa_N), \qquad \kappa_N \;\le\; C' \sqrt{\frac{b^D \, \log(bQD/\eta)}{N}},$$

with C, C' absolute constants. The proof is complete modulo one concentration estimate — the high-probability form of an ℓ_4-norm propagation step (Appendix L, Lemma L.7) — which we state precisely and leave as the single explicit gap; its expectation form is proved, and the theorem's multiplicative-error signature is confirmed in simulation (relative spread of m flat in D while the absolute spread shrinks ≈370× from D=1 to D=10).

Corollary 3.4 (Resolvable depth)**.** *Against the primitive-crosstalk noise floor $\nu(N) \propto \sqrt{1/N}$ of any linear readout, binding at depth D is resolvable when $2b^{-D}(1-\kappa_N) > \nu(N)$ (depth 1 is covered by Theorem E.7), i.e.*

$$D \;\le\; D^*(N) \;=\; \frac{\log N}{2 \log b} + O(1),$$

a staircase gaining one level per b^2-fold increase of N.

Simulation confirms the law exactly: across $b \in \{2, 3\}$ ($b = 2$ to depth 10, $b = 3$ to 7) and N from 256 to 65,536 (24 seeds per configuration; shared roles, with a side-by-side fresh-role sweep agreeing away from the aliased cells, Appendix L), the products $m \cdot b^D$ sit at 2 and 1 — within 0.014 and 0.006 at $N = 65{,}536$ — with the smaller-N deviations following Theorem 3.3's finite-N term, and the measured D^* staircase is exactly $1, 2, 3, 4, 5$ at $N = 256, \dots, 65{,}536$ for $b = 2$ (Figure 2; full tables in Appendix L). At CLIP's $N = 512$: the pure crossing $\log N/(2\log b)$ gives 4.5 ($b = 2$) and 2.8 ($b = 3$), and the measured $10/\sqrt{N}$ floor gives $D^* = 2$ and 1 — single digits either way. **The ceiling sits at ordinary natural-language nesting depth**: beyond it, resolving composition requires structured (tree-side) scoring rather than a single pooled vector.

A second phenomenon goes beyond attenuation.

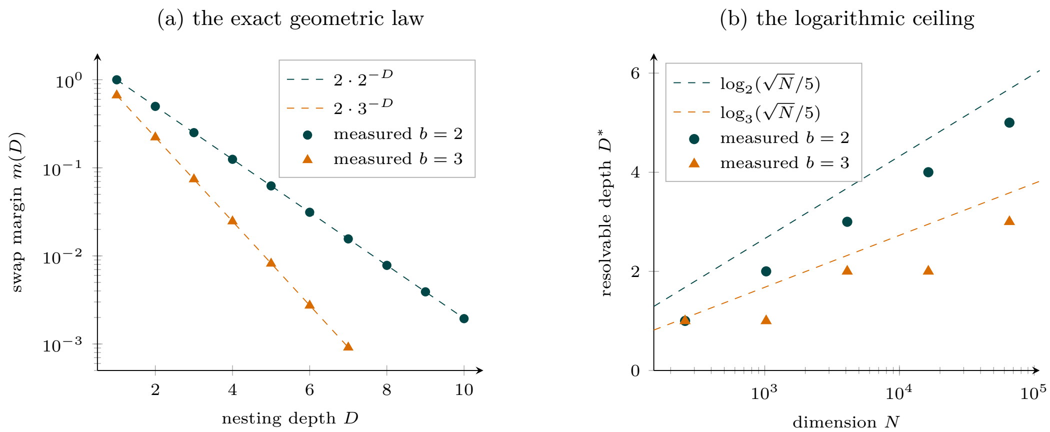


Figure 2: The depth ceiling, measured. **(a)** Swap margins sit on $2 \cdot b^{-D}$ across three orders of magnitude (log scale; $N = 65{,}536$ shown, where the products $m \cdot b^D$ are within 0.014 of 2; at smaller N the deviations grow as the finite-N term of Theorem 3.3 allows — Appendix L.4). **(b)** Resolvable depth (largest D with margin above the $10/\sqrt{N}$ floor) follows $\lfloor \log_b(\sqrt{N}/5) \rfloor$ exactly, including the integer staircase — the $D^* = \Theta(\log N / \log b)$ ceiling of Corollary 3.4.

Proposition 3.5 (Aliasing; Appendix L, Proposition L.12)**.** *With roles shared across levels, sign roles are self-inverse ($r \odot r = \mathbf{1}$), so a leaf's coefficient in the root expansion depends only on the* parity profile *of its role chain — which roles appear an odd number of times along its path. Leaves with equal parity profiles are exactly interchangeable: for unnormalized codes the swap margin is identically zero at any dimension N, with no orthogonality used; under per-node normalization the residue is crosstalk-sized, carrying no $2b^{-D}$ term.*

The simulation shows exactly this: aliased far swaps sit two to five orders of magnitude below the law, while fresh per-level roles restore $m \cdot b^D \approx 2.00$ on the same configurations. Flat self-inverse binding does not merely attenuate deep structure — it aliases some of it.

Scope. As a no-go this is class-relative: it binds the recursive role-binding class, not arbitrary encoders — an unconstrained encoder can look up any finite set of deep captions. The escape routes are structured (tree-side) scoring and chunking with cleanup, exactly the mechanisms of the VSA literature [10, 12]; on the vision–language side, text encoders that score along the syntactic derivation are a concrete instance of the tree-side route [27].

4 What the objective pays for binding

The second obstruction is about the training signal, and is architecture-free. A *contrastive sample* is a positive $(I_s, c(s))$ together with a negative caption c^-: with probability π the *explicit swap* $c(\sigma s)$, otherwise the caption of an independent scene. Writing $[c]$ for the *symbol class* of a caption — all captions sharing its symbol multiset — the *implicit collision rate* π_{impl} is the chance a random negative lands in the positive's class; at web scale both rates are vanishing. The loss is the one-negative logistic contrastive loss, over *arbitrary measurable scorers S* — no architecture assumption, so every bound applies to dual encoders a fortiori:

$$L(S) \;=\; \mathbb{E}\,\ell\big(S(I_s, c(s)) - S(I_s, c^-)\big), \qquad \ell(m) = \log(1 + e^{-m}).$$

Two comparators attach to any scorer: its *pairing-blind average* $\bar{S}$ (replace $S(I, c)$ by its average over $[c]$ — a bag-of-words scorer) and its *reflection* ρS (score every caption as if swapped — binding exactly reversed); $M_S(s)$ is the scorer's swap margin. Two regularity parameters: Δ-*separation* — cross-class contrasts are easy, $S(I_s, c(s)) - S(I_s, c') \geq \Delta$ whenever $[c'] \neq [c(s)]$ — and *within-class spread* R: $|S(I, c) - S(I, c')| \leq R$ for $c' \in [c]$. One structural fact matters for interpretation: with unrestricted capacity the population optimum *does* bind (Appendix K); the theorems say its advantage over not binding is a sliver. We prove:

Theorem 4.1 (The throttle; Appendix K, Theorem K.1 and Corollary K.2)**.** $L(\bar{S}) - L(S) \leq (\pi + \pi_{\mathrm{impl}}) \log 2 + e^{-(\Delta - 2R)}$. *Consequently the pairing-blind competitor is ε-optimal with $\varepsilon \leq (\pi + \pi_{\mathrm{impl}}) \log 2$ plus an exponential tail:* the objective's entire reward for binding — everything beyond bag-of-words — is confined to a sliver of that width. *No optimization guarantee weaker than the sliver can imply binding.*

Theorem 4.2 (Anti-binding is cheap; Appendix K, Theorem K.4)**.** $\left|L(\rho S) - L(S) - \pi\, \mathbb{E}_s[M_S(s)]\right| \leq \pi_{\mathrm{impl}} R + 2e^{-(\Delta - 2R)}$*: exactly reversed binding costs precisely* $\pi\, \mathbb{E}[M]$.

Both constants are verified to within 0.006 in controlled simulation (Appendix K), and three phenomena measured in the companion study [6] become corollaries: the dose–response of binding in π (linear at small π, slope $\mathbb{E}[M]$); the equivalence of up-weighting matched terms and raising π; and *cheap below-chance binding* — the reversed scorer sits only $\pi\, \mathbb{E}[M]$ away in loss, so small perturbations (the pooled code's cross-term interference — *leakage*, Appendix E — or implicit bias) suffice to select it, consistent with the backwards-binding observed in learned pooled codes (Appendix E).

5 The smoothness–binding frontier, and where deployed models sit

The third obstruction trades binding against the *similarity* job that deployment demands: paraphrases and order variants of a caption must embed close together. The two demands collide at the vocabulary level: an order variant of $c(s)$ and the swap $c(\sigma s)$ contain *identical words*, so any word-multiset-invariant text encoder performs the required merge and the forbidden one simultaneously. Formally:

Theorem 5.1 (Pairing-blind collapse; Appendix D, Theorem D.2)**.** *If the caption map depends only on the symbol multiset, then $\mathbf{t}(c(\sigma s)) = \mathbf{t}(c(s))$ and $M(s) = 0$ identically: the swap instances of (C_γ) fail for every $\gamma > 0$ and every image map, and swap benchmark accuracy is chance in expectation under any label-independent tie-breaking.*

Quantitatively, say the caption map is *δ-smooth* at s if both $\mathbf{t}(c(s))$ and $\mathbf{t}(c(\sigma s))$ have inner product at least $1 - \delta$ with some *pairing-neutral anchor* $\bar{t}_s$ — a fixed vector computable from the symbol multiset alone, e.g. the embedding of the *atom-list caption*, which lists the objects and attributes without pairing them (the anchor quantifier is what guards the bound from tautology; Appendix G).

Theorem 5.2 (Smoothness–binding frontier; Appendix G, Theorem G.3)**.** *If the caption map is δ-smooth at s, then for every image map, every dimension, and every training procedure,*

$$M(s) \;\leq\; \|\mathbf{t}(c(s)) - \mathbf{t}(c(\sigma s))\| \;\leq\; 2\sqrt{2\delta},$$

and the bound is tight: unit configurations achieve $2\sqrt{2\delta}\,(1 - O(\delta))$, so the rate $\sqrt{\delta}$ and the constant $2\sqrt{2}$ are exact.

The anchor-free inner inequality $|M(s)| \leq d(s) := \|\mathbf{t}(c(s)) - \mathbf{t}(c(\sigma s))\|$ is a per-item ceiling, and the theorem induces a text-only diagnostic: measure d, the smoothness gap δ realized against the atom-list anchor, the ceiling $\mathrm{cap} = 2\sqrt{2\delta}$, and the *frontier usage* $d/\mathrm{cap} \in [0, 1]$. Usage near 1 means binding is smoothness-limited — further gains must be paid for in paraphrase geometry; usage far below 1 means the ceiling is not the binding constraint.

We measured the diagnostic on the text encoders of 18 deployed models — CLIP variants across architectures and pretraining corpora, SigLIP [30], ALIGN [31], and the compositionality fine-tunes NegCLIP [2], SVLC [25], and DAC [26] — on a fixed grid of 675 *quartets*: a quartet fixes two objects and two color attributes, determining a scene and its swap; every model receives the identical grid and anchor template (protocol and rigor checks in Appendices H–I). Table 1 reports the grid aggregates, and Figure 3 plots them: every model sits at *aggregate usage* 0.248–0.349, a factor of three or more below the ceiling for all but NegCLIP (which comes closest, at 2.9×) — so binding failure in the deployed models is *not* a smoothness limit; by the throttle and the code-structure results it is an incentive and code limit. How models move within the figure's wedge is informative: three of the four

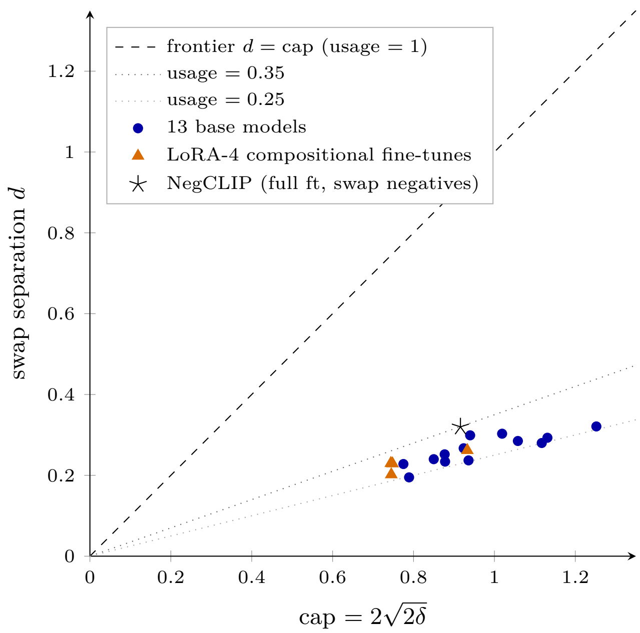


Figure 3: The model zoo against the frontier (the data of Table 1; axes to equal scale). Every deployed model sits in the narrow wedge between usage ≈ 0.25 and 0.35, far below the d = cap line: no model is smoothness-limited. Scale, data, and objective move points *along* the wedge; only NegCLIP (star) climbs *across* it, by raising d at roughly constant cap.

LoRA-rank-4 compositional fine-tunes move *along* the wedge (spending smoothness, not gaining separation; DAC-LLM raises d modestly), while NegCLIP — a full fine-tune with caption-level swap negatives, i.e. raised π — is the only model that climbs *across* it (usage 0.349). And the per-item ceiling is predictive on a real benchmark: across 14 models × 4,757 SugarCrepe items (66,598 decisions), subset-level mean d tracks accuracy at $r = 0.99$, with the swap subsets lowest exactly as the theory orders them (Appendix I).

6 What does not explain the failure

Two natural explanations are eliminated. *Dimension.* The axioms are satisfiable, and the threshold is known exactly:

Proposition 6.1 (Dimension dichotomy; Appendix C, Proposition C.3)**.** *For a concept system with at least $N + 2$ objects, the constituent-ranking axioms for all scene sizes $j \le k$ are satisfiable on $\mathbb{S}^{N-1}$ iff $k \le \lfloor N/2 \rfloor$, and in that regime placements exist satisfying margin retrieval (C_γ) as well, for some $\gamma > 0$.*

The mathematics is published, twice independently [7, 8]; random factored sign codes realize the possibility direction cheaply at deployed scales (Appendix C). So no dimension bound at deployed scales explains swap failure. *Exposure-graph connectivity.* The *exposure graph* is the bipartite graph recording which object–attribute pairs occur in training; it is tempting to conjecture that binding cannot generalize across its disconnected components (the gauge ambiguity of rank-one completion [15] — codes are identified only up to a per-component sign — supports this for learners that commit to a sign). The conjecture is false as an information bound: a sign-agnostic matcher, scoring candidates by the *absolute value* of the inner product, achieves accuracy 1.000 on cross-component swap tests even with zero same-component test pairs; coverage — how many pairs have been seen — not connectivity, is the threshold (Appendix M). Learners that do commit to a per-component sign drop below chance (0.27–0.37) — matching the below-chance binding that the throttle predicts is cheap.

Table 1: The frontier diagnostic across the text encoders of 18 deployed models: swap separation d, smoothness gap δ, ceiling cap $= 2\sqrt{2\delta}$, usage $= d/\text{cap}$, and the largest per-quartet usage; rows sorted by usage. Model references: [1, 29, 31, 30, 33, 32, 2, 25, 26]. Protocol, abbreviations, and rigor checks in Appendix H.

Model	Family / objective	d	δ	cap	usage	max
ALIGN-base	ALIGN (BERT text)	0.195	0.080	0.789	0.248	0.54
CLIP L/14-336	OpenAI CLIP	0.280	0.158	1.117	0.251	0.39
CLIP B/32	OpenAI CLIP	0.237	0.112	0.936	0.253	0.42
SigLIP-large	SigLIP (sigmoid)	0.321	0.199	1.252	0.256	0.55
CLIP L/14	OpenAI CLIP	0.293	0.162	1.131	0.259	0.41
MetaCLIP B/16	MetaCLIP	0.234	0.098	0.878	0.267	0.50
OpenCLIP H/14	LAION-2B	0.285	0.141	1.058	0.270	0.46
SVLC LLM+RB	CC3M, LoRA-4 ft	0.202	0.070	0.745	0.270	0.47
DAC-LLM	CC3M, LoRA-4 ft	0.262	0.110	0.933	0.281	0.48
CLIP B/16	OpenAI CLIP	0.240	0.091	0.850	0.283	0.49
OpenCLIP B/32	LAION-2B	0.252	0.097	0.877	0.287	0.52
OpenCLIP L/14	LAION-2B	0.267	0.108	0.924	0.289	0.44
MetaCLIP B/32	MetaCLIP	0.228	0.076	0.775	0.294	0.51
SigLIP-base	SigLIP (sigmoid)	0.303	0.132	1.019	0.297	0.55
SVLC RB	CC3M, LoRA-4 ft	0.229	0.071	0.748	0.306	0.51
DAC-SAM	CC3M, LoRA-4 ft	0.230	0.070	0.744	0.309	0.47
SigLIP2-base	SigLIP2 (sigmoid)	0.299	0.112	0.940	0.318	0.64
NegCLIP	COCO full ft, swap negs	**0.320**	0.107	0.916	**0.349**	0.57
13 base models: mean 0.275, median 0.270					0.248–0.318	

7 Related work

Kang et al. [5] initiated the ideal-encoder program; we formalize and extend it (Appendix B). Dimension/capacity limits for retrieval are characterized in the LIMIT line [21] and its possibility counterpart [8] — an orthogonal axis: those bounds vanish as N grows, while the frontier of §5 does not involve N, and the depth ceiling of §3 *grows* only logarithmically in it. Finite semantic resolution limits identification under non-compositional representations [22], without swap/margin content; Chen et al. [23] prove existence of swap-insensitive contrastive optima (existence only — our throttle quantifies the price and its reversal); SugarCrepe++ [20] documents the paraphrase/hard-negative dissociation that Theorem 5.2 formalizes. For depth, the sequence-memory theory of Frady et al. [12] supplies closely related crosstalk accounting, but no per-level margin law and no discrimination between a structure and its minimal edit — the gap our Theorems 3.2–3.3 close; Plate's classical discussion [10] is qualitative and centers the chunking escape. Compositional-generalization theory [13, 14, 16] concerns learners and data; our connectivity refutation (§6) sharpens which graph property carries the information. On the constructive side, compositional tensor-network encoders for vision–language — scoring along the syntactic derivation [27], and tensor-based compositional semantics for concept generalization [28] — implement the structured escape that the depth analysis motivates. Full positioning is in Appendix J.

8 Discussion

What the results support. The contingent causes of today's failures — vanishing π (the throttle), the pooled code's leakage (Appendix E), and the training distribution — are all fixable in principle, and the zoo measurements show the fixes moving models exactly as predicted (only raised π climbs the wedge). Behind them stand limits that survive all fixes: the smoothness–binding frontier (with measured slack today), and the depth ceiling $D^* = \Theta(\log N/\log b)$, which sits at natural-language depth for CLIP-sized models and binds the recursive role-binding class no matter the training.

Limitations. Our no-gos are class- or hypothesis-relative by necessity — the possibility theorem forbids anything stronger. The depth theorem's finite-N form rests on one explicitly flagged concentration estimate (its expectation form is proved, and the multiplicative-error signature is confirmed in simulation); the frontier measurement instantiates two-object color binding, not relations or counting; and the NegCLIP comparison carries a three-way confound (full fine-tune, corpus, negatives) that a rank-matched control would break. Open problems, including the margin-robust dimension question and the naturalistic frontier, are collected in Appendices J and N.

Reproducibility. Every measured claim carries its protocol in the appendices; measurement scripts and raw results are available from the author on request.

Appendix

The appendices are a self-contained technical development: complete definitions and axioms, all proofs, measurement protocols with their raw numbers, and negative results. Results stated in the main text reappear here inside their full development, with the same notation. One proof gap is left deliberately explicit: the concentration estimate inside Lemma L.7 (Appendix L), which is the single unproved step behind Theorem L.8.

A Notation and glossary

Throughout, $\mathbb{S}^{N-1} = \{x \in \mathbb{R}^N : \|x\| = 1\}$ is the unit sphere in $\mathbb{R}^N$; $\langle \cdot, \cdot \rangle$ is the Euclidean inner product and $\|\cdot\|$ the Euclidean norm. For unit vectors the inner product equals the cosine similarity. We use c, c_1, c_2 for absolute constants and $O(\cdot)$, $\Theta(\cdot)$ with their usual meaning.

Symbol	Name	Meaning (first defined in)
O, A	objects, attributes	finite sets, $\|O\| = M$, $\|A\| = K$ (§B)
s	scene	a set of k object–attribute pairs with distinct objects (§B)
$c(s), I_s$	caption, image	canonical linguistic/visual realization of s (§B)
$\mathcal{C}(s), \mathcal{I}(s)$	realization classes	all captions / all depictions of s (§B)
σs	swap	the scene with two attributes exchanged (§B)
$\mathcal{N}_{\text{full}}(s), \mathcal{N}_1(s)$	negative families	all other scenes' captions; hard core = one edit (replace / add–drop / swap) (§B)
$\mathbf{i}, \mathbf{t}$	encoders	image/text embedding maps into $\mathbb{S}^{N-1}$ (§B)
$M(s)$	binding margin	$\langle \mathbf{i}(I_s), \mathbf{t}(c(s)) - \mathbf{t}(c(\sigma s)) \rangle$ (§B)
$d(s)$	swap separation	$\|\mathbf{t}(c(s)) - \mathbf{t}(c(\sigma s))\|$ (§G)
$\bar{t}_s$	pairing-neutral anchor	a unit vector depending only on the symbol multiset of s (§G)
δ	smoothness gap	both pairings within cosine $1 - \delta$ of $\bar{t}_s$ (§G)
cap	frontier ceiling	$2\sqrt{2\delta}$ (§G)
usage	frontier position	$d/\text{cap} \in [0, 1]$ (§G)
g, ℓ, ℓ_0, L	gain, leakage, cross-talk	code-quality functionals for pooled codes (§E)
π	swap-negative rate	probability the training negative is a swap (§F)

B The formal setup

B.1 Concept systems, scenes, and negatives

Definition B.1 (Concept system and scenes)**.** Fix finite sets O (*objects*, $|O| = M$) and A (*attributes*, $|A| = K$). An *atom* is a pair $(o, a) \in O \times A$, read "object o has attribute a." A k-*scene* is a set $s = \{(o_1, a_1), \ldots, (o_k, a_k)\}$ with the o_j pairwise distinct. Where a swap is discussed we additionally require the a_j pairwise distinct (otherwise the swap can be the identity). We write $V(s) = \{o_1, \ldots, o_k\}$ for the object set of s. Each scene has a *caption* $c(s)$ (a linguistic description asserting exactly the atoms of s) and an *image* I_s (a depiction of exactly those atoms). Example with $k = 2$: $s = \{(\text{car}, \text{red}), (\text{dog}, \text{blue})\}$, $c(s) =$ "a red car and a blue dog." The *symbol multiset* of s is the multiset of object and attribute symbols occurring in s *with the pairing forgotten* (here: {car, dog, red, blue}). Atoms are bound pairs, so a scene and its swap (below) have *different* atom sets but the *same* symbol multiset; captions realize the symbol multiset as a word multiset.

Remark B.2 (Why $c(s)$ but I_s). The asymmetric notation is deliberate. The caption constructor is used *as a function* throughout: the axioms apply it to *edited* scenes ($c(\sigma s)$ below, and every hard negative is the caption of an edited scene)—including scenes that are never depicted; a false caption needs no image to exist. The image realization, by contrast, is never applied to an edited scene: no axiom will mention $I_{\sigma s}$. So $c(\cdot)$ is an operator composed with scene transformations, while I_s labels a given datum. This mirrors the benchmarks themselves, whose negatives are synthesized on the text side.

Definition B.3 (Swap and the negative families)**.** For a scene s with $k \geq 2$ and two chosen positions, the *swap* σs is the scene with the two attributes exchanged: from

$\{(o_1, a_1), (o_2, a_2), \dots\}$ to $\{(o_1, a_2), (o_2, a_1), \dots\}$ — a single *transposition* of the pairing (general permutations are compositions of transpositions; see Remark B.4). Note $c(s)$ and $c(\sigma s)$ contain the *same multiset of words*; they differ only in which attribute is bound to which object.

Two negative families are distinguished. The *full family* $\mathcal{N}_{\text{full}}(s)$ consists of *every caption that describes a scene other than* s (formalized by realization classes in Definition B.5). The *hard core* $\mathcal{N}_1(s) \subset \mathcal{N}_{\text{full}}(s)$ consists of the captions of scenes *one edit away* from s: (i) *replace* one atom's object or attribute by one not occurring in s; (ii) *add* an atom whose object does not occur in s, or *drop* one atom; (iii) *swap*: one transposition of the pairing. The axiom (C_γ) below quantifies over the full family; the hard core is the binding-critical worst case and the part that benchmarks instantiate (Remark B.4).

The hard core maps onto the hard-negative benchmarks as follows. SugarCrepe's seven subsets are exactly REPLACE-object/attribute/relation, ADD-object/attribute, and SWAP-object/attribute [3]: its taxonomy coincides with (i)–(iii) minus *drop* (on which see Remark B.4). The swap family (iii) is the canonical content of ARO [2]: VG-Attribution exchanges the two attributes between two objects, and VG-Relation exchanges the two arguments of a relation. Winoground [4] realizes pairing permutations with *identical word multisets* (each caption pair contains the same words, differently bound)—i.e. swap-type negatives in our sense, though a fraction of its items are known to require capabilities beyond pairing [17]. One benchmark family is deliberately *absent* from $\mathcal{N}_{\text{full}}(s)$: ARO's COCO-Order/Flickr-Order subsets, whose negatives are ungrammatical word *shuffles*; those are detectable from the text alone, with no image—the "hackability" SugarCrepe was constructed to remove [3]—and correspond to no scene transformation. This identification is what makes the axioms below a formalization of "benchmark competence" rather than an arbitrary list of geometric conditions.

Remark B.4 (What exactly is a negative). Four clarifications fix the ambiguities latent in "negative." (1) *Retrieval, not falsity.* (C_γ) is an argmax-with-margin condition: given I_s, the *exact* description $c(s)$ must out-score every caption of every other scene. Negatives need not be false in the natural-language sense; they need to describe a different scene. This matters for *drop*: under the closed-world convention of Definition B.1 (captions assert their atoms *exactly*), the dropped caption "a red car" describes the one-object scene, which I_s does not depict, so preferring the full caption is a completeness-of-description demand. Under open-world semantics a dropped caption is simply *true*, which is why no benchmark tests drops (SugarCrepe has no drop subset). Our zoo probe (§H) uses swaps; the benchmark protocol of §I uses the five swap/replace subsets (the two add subsets would serve equally—their omission is scoping, not principle). Drop is an axiom-internal negative, not an empirical one. (2) *Why the hard core is the right worst case.* Within the pooled class of §E, margins are monotone in edit distance: single-add $\approx \frac{1}{2(k+1)} <$ single-replace $\approx \frac{1}{k} <$ swap $\approx \frac{2}{k} <$ multi-edit negatives (an m-edit difference has m matched terms and margin $\approx m/k$ by an outlined extension of the route of Theorem E.7). The minimum over $\mathcal{N}_{\text{full}}$ is attained on $\mathcal{N}_1$, so for such codes proving the hard core proves the family. *Caveat stated plainly:* for *arbitrary* encoders, hard-core margin implies nothing about further captions — which is exactly why the axiom quantifies over $\mathcal{N}_{\text{full}}$. No theorem is weakened: Theorems D.2 and G.3 use only the swap instance, Theorem E.7 proves the hard core and inherits the rest by the monotonicity above, and the possibility construction of Proposition C.3(ii) satisfies the full family outright ($\langle \mathbf{i}(I_s), \mathbf{t}(c(s)) \rangle = 1$ beats every distinct unit vector). (3) *Freshness.* Replacement atoms must not occur in s: for objects this is forced (scenes have distinct objects), for attributes it preserves the distinct-attribute convention. (4) *Swaps are transpositions.* A general permutation of the pairing rebinds ≥ 3 atoms and, in the pooled class, separates *more* than a transposition ($\approx m/k$ for an m-cycle); the transposition is the binding-critical case, and it is what ARO/SugarCrepe instantiate.

Definition B.5 (Realization classes)**.** A scene admits many linguistic and visual realizations. Write $\mathcal{C}(s)$ for the set of captions asserting exactly the atoms of s—order variants ("a red car and a blue dog" / "a blue dog and a red car"), determiner and phrasing variants, paraphrases—and $\mathcal{I}(s)$ for the set of depictions of exactly s. Since captions assert their atoms exactly, the classes $\mathcal{C}(s)$ are pairwise disjoint across scenes; in particular the swap caption is *not* an order variant: it lies in the different class $\mathcal{C}(\sigma s)$. We fix canonical representatives $c(s) \in \mathcal{C}(s)$ and $I_s \in \mathcal{I}(s)$, and adopt the convention that every axiom instance quantifies over realizations: (C_γ) reads "for all $I \in \mathcal{I}(s)$, all $c^+ \in \mathcal{C}(s)$, and all $c' \in \mathcal{N}_{\text{full}}(s) := \bigcup_{s' \neq s} \mathcal{C}(s')$" (the hard core $\mathcal{N}_1(s)$ is the union over one-edit scenes only). Every theorem below holds verbatim for each choice of realizations—each proof manipulates one caption pair and one image at a time—so statements are written with representatives; the class structure is revisited where it carries content (Remark B.6, Corollary G.5). Image multiplicity, in particular, changes nothing anywhere: all bounds are per-image, and the constructions place images inside open cones with room for any number of them.

Remark B.6 (Required vs. forbidden invariance collide on the vocabulary). The similarity job (a deployment demand, formalized in §G; not an axiom of Definition B.8) requires $\mathbf{t}$ to (approximately) identify $\mathcal{C}(s)$—in particular its order variants; the binding job requires $\mathbf{t}$ to separate $\mathcal{C}(s)$ from $\mathcal{C}(\sigma s)$. At the level of *word multisets* these demands are indistinguishable: the order variant "a blue dog and a red car" $\in \mathcal{C}(s)$ and the swap "a blue car and a red dog" $\in \mathcal{C}(\sigma s)$ contain the identical words; they differ only in syntactic attachment. Consequently (a) any word-multiset-invariant (bag-of-words) encoder performs the required conflation and the forbidden one *simultaneously*—Theorem D.2 is the formal statement; (b) any encoder that separates swaps must compute something about the *parse*, not the vocabulary. This is the sharpest form we know of "binding requires syntax," and it is visible only once realization classes are explicit; Figure 5 draws the collision.

B.2 Dual encoders and the axioms

Definition B.7 (Dual encoder)**.** A *dual encoder* is a pair of injective maps $\mathbf{i} : \{\text{images}\} \to \mathbb{S}^{N-1}$, $\mathbf{t} : \{\text{captions}\} \to \mathbb{S}^{N-1}$. The *score* of an image–caption pair is $\langle \mathbf{i}(I), \mathbf{t}(c) \rangle$ (cosine similarity; this is the deployed scoring rule of CLIP-class models). We first treat $\mathbf{i}, \mathbf{t}$ as *placements*—arbitrary assignments of unit vectors, the "oracle" reading of Kang et al. [5]—and only in §F as maps produced by training.

Definition B.8 (The axioms)**.** Fix a margin $\gamma > 0$ and a maximal scene size $k_{\max}$. Convention: throughout the document k denotes the size of the scene at hand ($k = |s|$); the axioms quantify over all scenes with $k \leq k_{\max}$.

(C_γ) **Margin retrieval.** For every scene s with at most $k_{\max}$ objects and every $c' \in \mathcal{N}_{\text{full}}(s)$, the full negative family of Definition B.3:

$$\langle \mathbf{i}(I_s),\, \mathbf{t}(c(s)) \rangle \;\geq\; \langle \mathbf{i}(I_s),\, \mathbf{t}(c') \rangle + \gamma.$$

(K_k) **Constituent ranking.** For every k-scene s, every $x \in V(s)$ and every $z \in O \setminus V(s)$:

$$\langle \mathbf{i}(I_s),\, \mathbf{t}(x) \rangle \;>\; \langle \mathbf{i}(I_s),\, \mathbf{t}(z) \rangle,$$

where $\mathbf{t}(x)$ denotes the embedding of the single-object caption for x (the caption domain includes such object-only captions; $\mathbf{t}$ is defined on all of them).

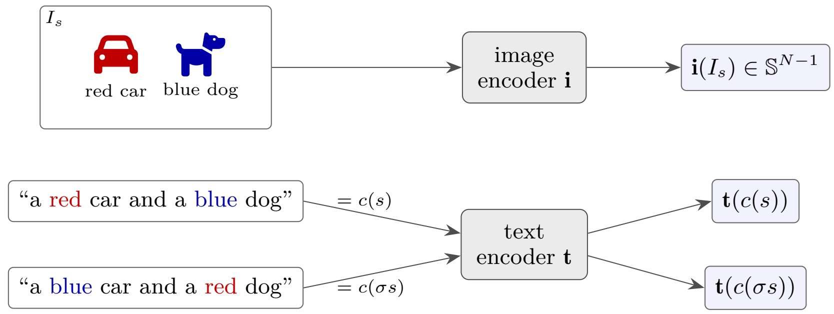
(C_γ) says the image of a scene prefers its own caption to every hard negative, with margin. (K_k) says the image of a scene ranks all objects *present* above all objects *absent*—the "does the embedding know what is in the picture" condition.

Definition B.9 (Binding margin)**.** For a scene s and its swap, the *binding margin* is

$$M(s) \;:=\; \langle \mathbf{i}(I_s),\; \mathbf{t}(c(s)) - \mathbf{t}(c(\sigma s)) \rangle.$$

The swap instance of (C_γ) is precisely $M(s) \geq \gamma$. Benchmark accuracy on a swap item is $\mathbf{1}[M(s) > 0]$; $M(s)$ is the quantity every theorem below controls. Figure 4 shows the whole pipeline on the running example.

Remark B.10 (Relation to Kang et al.). Kang et al. [5] formulate ideal-encoder conditions closely related to Definition B.8 and derive from them an impossibility claim for binding. Our formulation differs in three presentational respects, discussed further in Paper 1 (§3): every condition is quantified explicitly over scenes and realization classes; the condition list is organized around the single margin family (C_γ), since a margin subsumes the various distinctness conditions in one clause; and a uniform convention for strict versus weak inequalities is fixed, which matters at the collapse point. We retain their Condition 1.1 as (K_k), which carries the interesting geometry. Every theorem below is proved from Definition B.8 directly, so the development is self-contained.



binding margin $\quad M(s) = \langle \mathbf{i}(I_s), \mathbf{t}(c(s)) - \mathbf{t}(c(\sigma s)) \rangle$; $\quad$ the benchmark item is correct $\iff M(s) > 0$.

Figure 4: The dual-encoder binding problem (Definitions B.1–B.9). One image of the scene s; two captions with the *identical word multiset*, differing only in which attribute is bound to which object. Both towers embed into the unit sphere, and the deployed score is a single inner product. Every theorem in this paper is a statement about the margin $M(s)$.

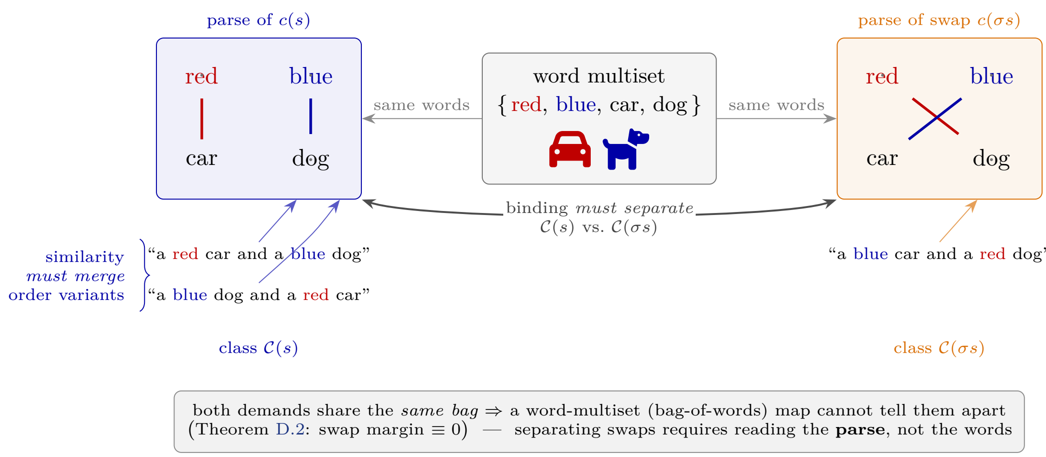


Figure 5: Same vocabulary, two parses (Remark B.6). An order variant of $c(s)$ and the swap $c(\sigma s)$ share the identical word multiset; they differ only in the binding (which colour attaches to which object). The invariance similarity *requires* —merge the order variants of $\mathcal{C}(s)$ (left)—and the invariance binding *forbids*—separate the swap class $\mathcal{C}(\sigma s)$ (right)—are indistinguishable at the vocabulary level. A bag-of-words map performs both at once (Theorem D.2); separating swaps requires reading the parse.

C Possibility: the axioms have solutions

Before proving impossibility theorems one must know what is possible; otherwise an "impossibility" may simply be hiding in an inconsistent axiom system. The content of this section is a dichotomy for (K_k), which turns out to be *published*—twice, independently—so we state it as a cited proposition, prove the short direction (the impossibility) in full because it is used later, and outline the construction. The section's purpose in the larger argument is its Corollary C.4: *the full axiom system, including all swap conditions, is satisfiable with a positive margin.* Hence every impossibility theorem in this paper must—and does—enter through an explicitly added hypothesis.

C.1 Convex-geometry vocabulary

Definition C.1. Let $X \subset \mathbb{R}^N$ be finite. $\operatorname{conv}(X)$ is the convex hull. A point $x \in X$ is *extreme* if $x \notin \operatorname{conv}(X \setminus \{x\})$. A subset $T \subseteq X$ is *(strictly linearly) separable* from $X \setminus T$ if

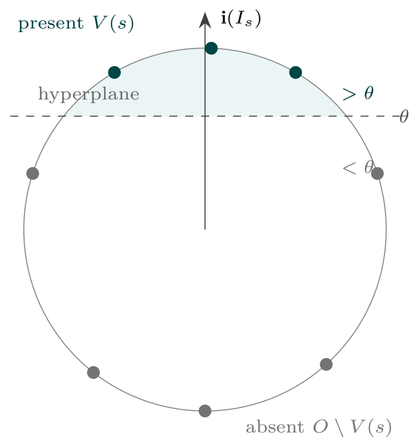


Figure 6: What (K_k) demands, for one scene (the reduction opening this subsection). The single-object embeddings $\{\mathbf{t}(x)\}$ lie on the sphere; the scene's image vector $\mathbf{i}(I_s)$ acts as the linear functional $y \mapsto \langle \mathbf{i}(I_s), y \rangle$—"height along the arrow." Condition (K_k) requires this functional to rank every *present* object $x \in V(s)$ above every *absent* one, i.e. the dashed hyperplane $\langle \mathbf{i}(I_s), \cdot \rangle = \theta$ strictly separates $\{\mathbf{t}(x) : x \in V(s)\}$ (teal, in the shaded accept-region) from its complement (grey). So "(K_j) for all j-scenes" is exactly "every j-subset of the M points is linearly separable"—a demand Radon's theorem forbids once $j > \lfloor N/2 \rfloor$ (Figure 7 is the same picture when separation *fails*).

there is $w \in \mathbb{R}^N$ and $\theta \in \mathbb{R}$ with $\langle w, x \rangle > \theta$ for $x \in T$ and $\langle w, x \rangle < \theta$ for $x \in X \setminus T$. A *face* of the polytope $P = \operatorname{conv}(X)$ is the intersection of P with a supporting hyperplane. P is *k-neighborly* if every subset of at most k vertices spans a face. The *trigonometric moment curve* in $\mathbb{R}^{2m}$ is $\gamma(\theta) = (\cos\theta, \sin\theta, \cos 2\theta, \sin 2\theta, \ldots, \cos m\theta, \sin m\theta)$; it lies on the sphere of radius $\sqrt{m}$, and the convex hull of any finite set of distinct points on it is a *cyclic polytope*, which is m-neighborly [9].

One elementary fact we use twice:

Lemma C.2 (Sphere points are extreme)**.** *Distinct points on $\mathbb{S}^{N-1}$ are extreme points of their convex hull.*

Proof. If $x = \sum_i \lambda_i y_i$ with $y_i \in \mathbb{S}^{N-1}$, $\lambda_i > 0$, $\sum \lambda_i = 1$, then $1 = \|x\| \le \sum_i \lambda_i \|y_i\| = 1$, with equality in the triangle inequality only if all y_i are positive multiples of one another, hence (being unit) all equal to x. □

C.2 The dimension dichotomy

Observe that (K_k) constrains only the *single-object* embeddings $\{\mathbf{t}(x) : x \in O\}$: for the scene s, the linear functional $y \mapsto \langle \mathbf{i}(I_s), y \rangle$ must strictly separate $\{\mathbf{t}(x) : x \in V(s)\}$ from the embeddings of all absent objects. So (K_j) for all scenes of size j requires: *every j-subset of the M points $\{\mathbf{t}(x)\} \subset \mathbb{S}^{N-1}$ is strictly separable from its complement.* Figure 6 shows the demand on one scene; the dichotomy below asks for *which* (N, k) it can hold for *all* scenes at once.

Proposition C.3 (Dimension dichotomy)**.** *Let $M \ge N+2$ and consider the axioms (K_j) for all scene sizes $j \le k$ (the* cumulative *reading: a system competent on k-scenes is competent on smaller ones).*

(i) If $k > \lfloor N/2 \rfloor$, no placement of $\{\mathbf{t}(x)\}_{x \in O}$ on $\mathbb{S}^{N-1}$ satisfies them.

(ii) If $k \le \lfloor N/2 \rfloor$, placements exist satisfying (K_j) for all $j \le k$ and *(C_γ) for all scenes with at most k objects, for some $\gamma > 0$.*

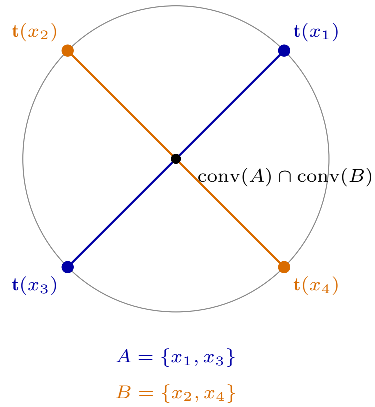


Figure 7: The Radon impossibility (Proposition C.3(i)) in its smallest instance: $N = 2$, $M = 4 = N + 2$ object embeddings on the circle $\mathbb{S}^1$. Radon's theorem partitions them into A and B whose hulls (the two chords) intersect; no linear functional is therefore constant-sign on $\{\mathbf{t}(x_1), \mathbf{t}(x_3)\}$ and opposite on the rest, so no image vector can rank the 2-scene $\{x_1, x_3\}$ above its complement: (K_2) fails. In general the smaller Radon side has at most $\lfloor N/2 \rfloor + 1$ points, giving the threshold $k \leq \lfloor N/2 \rfloor$.

Attribution. This dichotomy is published twice: as the *convex dimension of complete k-uniform hypergraphs* (Martínez-Sandoval and Padrol [7]: $\mathrm{cd}(K_n^{(k)}) = 2k$ for $n \geq 2k+2$, with the same Radon lower bound and neighborly upper bound), and as the *minimal embeddable dimension* for top-k retrieval ([8]: $\mathrm{MED}(m, k) = \min\{2k, m-1\}$ for linear scoring, their Prop. A.2; $\Theta(k)$ for inner-product/Euclidean/cosine in their main text). Neither literature cites the other; the bibliographic bridge appears to be new, the mathematics is not. We claim nothing here beyond the use we make of it.

Proof of (i). Radon's theorem states: any $N + 2$ points in $\mathbb{R}^N$ admit a partition into two nonempty sets A, B with $\mathrm{conv}(A) \cap \mathrm{conv}(B) \neq \emptyset$. Apply it to any $N+2$ of the M embedding points. By Lemma C.2, no single point lies in the hull of the others, so both Radon sides have at least 2 points. The smaller side A has $|A| \leq \lfloor (N+2)/2 \rfloor = \lfloor N/2 \rfloor + 1 \leq k$. Now consider any scene s with $V(s) = A$ (such a scene exists since $2 \leq |A| \leq k$). Strict separation of $\{\mathbf{t}(x) : x \in A\}$ from the rest is impossible: any functional constant-sign on A and opposite-sign on its complement would separate $\mathrm{conv}(A)$ from $\mathrm{conv}(\text{rest}) \supseteq \mathrm{conv}(B)$, but these intersect. So $(K_{|A|})$ fails. Figure 7 shows the smallest case. □

Proof sketch of (ii). Two steps. *Step 1 (single-object layer).* Cumulative separability of all $\leq k$-subsets is equivalent to k-neighborliness of the hull (for point sets in the regime $M \geq 2k+2$, which holds here since $M \geq N+2 \geq 2k+2$; the equivalence genuinely requires the cumulative quantifier: for 4 points on a circle *every* 3-subset is separable, since its complement is a single extreme point, yet *none* is a face; cumulative separability already fails at the 2-subsets, the Radon diagonals). Place the M object embeddings on the trigonometric moment curve: for even N its cyclic polytopes are $\lfloor N/2 \rfloor$-neighborly and inscribed; for odd N use the construction in an equatorial $\mathbb{S}^{N-2}$, which loses nothing because $\lfloor (N-1)/2 \rfloor = \lfloor N/2 \rfloor$ for odd N. Every $\leq k$-subset now spans a face, whose supporting functional can be perturbed to a strict separator with an open cone of witnesses. *Step 2 (composite layer).* For each scene s place $\mathbf{t}(c(s))$ at a generic point of the (open, nonempty) witness cone of $V(s)$, and set $\mathbf{i}(I_s) := \mathbf{t}(c(s))$ (cross-modal coincidence is permitted—injectivity is per-map; if undesired, perturb $\mathbf{i}(I_s)$ within the cone). Then (K_j) holds because $\mathbf{i}(I_s)$ is a separating witness for $V(s)$, and each (C_γ) instance holds because $\langle \mathbf{i}(I_s), \mathbf{t}(c(s)) \rangle = 1$ strictly exceeds the inner product with any *distinct* unit vector, including the swap caption; positivity of a uniform γ follows from finiteness. □

Corollary C.4 (No free-placement obstruction to binding)**.** *For $k \leq \lfloor N/2 \rfloor$ (e.g. any realistic caption length at $N = 512$), there exist placements satisfying every axiom of Definition B.8 with $\gamma > 0$. In particular the swap conditions—binding—are satisfiable. Any binding impossibility theorem on this setup must therefore introduce an additional hypothesis, and is exactly as strong as that hypothesis is natural.*

This corollary is central to the whole development. Where Kang et al. emphasize impossibility, the picture here is complementary: ideal CLIP is possible under the axioms alone, and impossibility appears exactly when one adds a structural, statistical, or semantic hypothesis. The next three sections add those hypotheses one at a time.

D Obstruction I: pairing-blind encoders

Definition D.1 (Pairing-blind maps)**.** A caption map $\mathbf{t}$ is *pairing-blind* (or *additive*) if $\mathbf{t}(c)$ depends only on the *symbol multiset* of the described scene (Definition B.1)— equivalently, on the word multiset of c—not on which attribute is bound to which object. The canonical example: any encoder whose caption representation is a sum or mean of per-*word* vectors (bag-of-words pooling). A sum of per-*atom* codes is *not* pairing-blind—atoms are bound pairs—and §E is built on exactly that distinction.

Theorem D.2 (Exact structural impossibility)**.** *If $\mathbf{t}$ is pairing-blind then for every scene s, $\mathbf{t}(c(\sigma s)) = \mathbf{t}(c(s))$, hence $M(s) = 0$ identically: the swap instances of (C_γ) fail for every $\gamma > 0$ and every image map, and swap benchmark accuracy is chance in expectation under any label-independent tie-breaking. For a pairing-blind* image *map the conclusion holds in pair form: $M(s) + M(\sigma s) = 0$, so the swap instances for s and σs cannot both hold (though a single one may).*

Proof. $c(s)$ and $c(\sigma s)$ share the same symbol multiset (Definition B.1), so a pairing-blind $\mathbf{t}$ maps them to the same point; the margin $M(s) = \langle \mathbf{i}(I_s), \mathbf{t}(c(s)) - \mathbf{t}(c(\sigma s))\rangle = \langle \mathbf{i}(I_s), 0\rangle = 0$. For a pairing-blind image map, $\mathbf{i}(I_s) = \mathbf{i}(I_{\sigma s})$, and $M(s) + M(\sigma s) = \langle \mathbf{i}(I_s), \mathbf{t}(c(s)) - \mathbf{t}(c(\sigma s))\rangle + \langle \mathbf{i}(I_s), \mathbf{t}(c(\sigma s)) - \mathbf{t}(c(s))\rangle = 0$. □

Three lines, drawn in Figure 8—and yet this is, we contend, the true core of the "CLIP cannot bind" phenomenon in the literature. Everything else in derivations of that flavor (sphere geometry, superposition lemmas, argmax selections) is scaffolding around the observation that *a multiset function cannot distinguish rearrangements*. The theorem also calibrates what an impossibility result must look like: it holds with no dimension assumption, no training assumption, and an *exact* conclusion—because the hypothesis (additivity) is maximally strong. The next section weakens the hypothesis to realistic compositional codes and watches the exact impossibility become a quantitative margin theory.

E Obstruction II: pooled codes and leakage

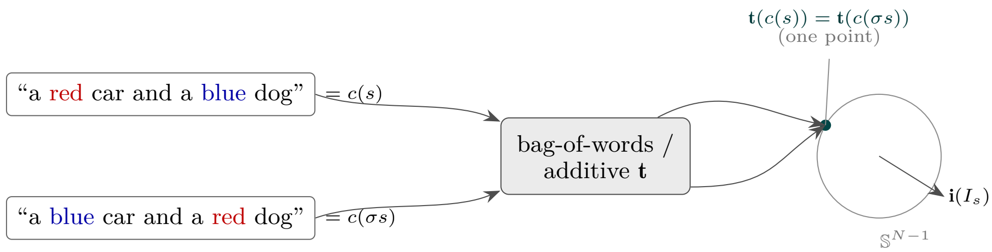
Real caption towers are neither free placements nor exactly additive: they compose *per-atom codes* into a single vector.

E.1 The homomorphism form: what "compositional" means here

The classical formalization of compositionality (Montague's) is a homomorphism condition:

$$(H) \qquad \mathbf{t}(\text{whole}) \;=\; \Phi\big(\{\mathbf{t}(\text{part}_i)\}_i\big)$$

for a fixed combination operator Φ (reserving g for the gain of Definition E.2): the representation of an expression is a function of the representations of its parts. Two calibrations before we use it. First, (H) has content only relative to a decomposition into parts that is *fixed in advance*: with Φ and the parts both free, every map is trivially compositional [18]; and real transformer encoders are contextual (the vector of "car" inside a caption is not a fixed $\mathbf{t}(\text{car})$), so (H) is an idealization—precisely the idealization the pooled codes below make exact. Second, the choice of parts is the whole game:



$$M(s) = \langle \mathbf{i}(I_s),\, \mathbf{t}(c(s)) - \mathbf{t}(c(\sigma s))\rangle = \langle \mathbf{i}(I_s),\, \mathbf{0}\rangle = 0$$

same word multiset ⇒ same point ⇒ zero margin, for *every* image encoder

Figure 8: Theorem D.2 (bag-of-words collapse). A pairing-blind text map sends both pairings to the *same* point—they share a word multiset—so the binding margin is exactly zero for every image encoder, dimension, and training. Contrast Figure 4, where the two captions are distinct points; the whole "CLIP cannot bind" phenomenon is this one observation that a multiset function cannot distinguish rearrangements.

- **Parts = words**, Φ symmetric. The scene and its swap present the identical word multiset, so (H) forces $\mathbf{t}(c(s)) = \mathbf{t}(c(\sigma s))$: this *is* pairing-blindness, and Theorem D.2 is the one-line corollary.
- **Parts = bound constituents** (the parse's phrases), via an inner $h(\mathbf{t}(\text{attribute}), \mathbf{t}(\text{object}))$ and an outer symmetric Φ (notation of the definitions below). The swap now changes *which* phrases exist, so the margin can survive. The pooled factored codes of this section are exactly this two-level homomorphism: $h = \odot$ on typed codebooks, $\Phi =$ normalized sum.

This restates "it is the code, not the pooling" algebraically: the outer pooling may be symmetric—what decides binding is whether the pooled units are words or *bracketed pairs*. Note also what need *not* be asymmetric: $\odot$ is commutative, role information is carried by the typed codebooks (u for objects, v for attributes), and the load-bearing structure is the bracketing (Figure 5). The recursion implicit in (H) also marks the one axis this paper leaves open—nesting depth (§J, open problems).

E.2 Pooled codes, gain, leakage

The natural intermediate class is:

Definition E.1 (Pooled codes)**.** A *pooled code* assigns to each atom (o, a) unit vectors $W_{\mathbf{i}}(o, a)$, $W_{\mathbf{t}}(o, a) \in \mathbb{R}^N$ (the *codebooks*) and embeds

$$\mathbf{i}(I_s) = \frac{S_{\mathbf{i}}}{\|S_{\mathbf{i}}\|}, \quad S_{\mathbf{i}} = \sum_{j=1}^{k} W_{\mathbf{i}}(o_j, a_j), \qquad \mathbf{t}(c) = \frac{S_{\mathbf{t}}}{\|S_{\mathbf{t}}\|}, \quad S_{\mathbf{t}} = \sum_{j=1}^{k} W_{\mathbf{t}}(o_j, a_j).$$

The code is *tied* if $W_{\mathbf{i}} = W_{\mathbf{t}} =: W$.

Definition E.2 (Gain, leakage, cross-talk)**.** For a pooled code define

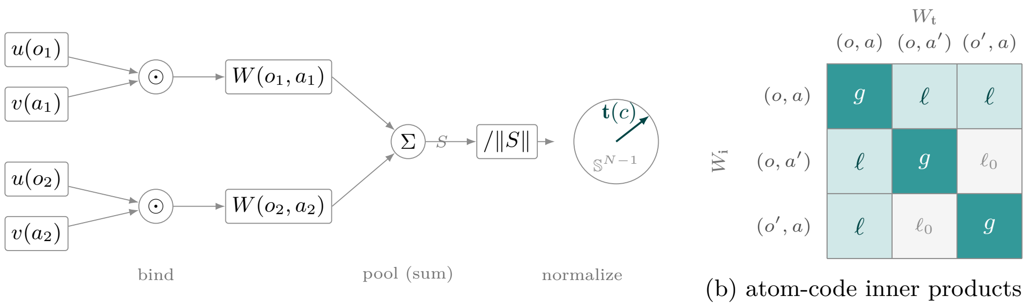
$$g := \langle W_{\mathbf{i}}(o, a),\, W_{\mathbf{t}}(o, a)\rangle \quad \text{(exact-match gain)},$$

$$\ell := \max_{\text{one shared atom}} \big|\langle W_{\mathbf{i}}(o, a),\, W_{\mathbf{t}}(o, a')\rangle\big| \vee \big|\langle W_{\mathbf{i}}(o, a),\, W_{\mathbf{t}}(o', a)\rangle\big| \quad \text{(leakage)},$$

$$\ell_0 := \max_{\text{no shared atom}} \big|\langle W_{\mathbf{i}}(o, a),\, W_{\mathbf{t}}(o', a')\rangle\big| \quad \text{(cross-talk)}, \qquad L := \max(\ell, \ell_0).$$

Leakage is the score a code grants a *partially* matching atom—right object, wrong attribute, or vice versa. Paper 1 identified accumulated leakage as the mechanism of the anti-generalization of learned MLP composition codes (trained codes develop $\ell \approx +0.17$ while random factored codes have $\ell \approx 0.00$); here it becomes the organizing functional of a theorem. Figure 9 draws the code (a) and the functional (b).

Throughout this section scenes have distinct objects *and* distinct attributes, and $k \geq 2$.



(a) bind → pool → normalize

(b) atom-code inner products $\langle W_{\mathrm{i}}, W_{\mathrm{t}} \rangle$

Figure 9: Pooled factored codes and leakage (Section E). **(a)** Each atom (o_j, a_j) is bound by a Hadamard product $u(o_j) \odot v(a_j)$; the bound codes are summed and normalized to the caption embedding $\mathbf{t}(c)$. **(b)** The code's quality is set by *leakage* ℓ — the inner product a *partial* match (right object, wrong attribute) receives — against the exact-match gain g. Random tied factored codes achieve $g = 1$, $\ell \approx 0$; additive (bag-of-words) codes sit at $\ell = \Theta(g)$ and cannot bind.

E.3 The norm-cancellation lemma

Margins of pooled codes are differences of two cosines whose numerators *and denominators* both change under the edit. Bounding the two norms independently is fatally lossy; the following trivial identity is what rescues the computation.

Lemma E.3 (Norm cancellation)**.** *Let $S' = S - D$. Then*

$$\|S'\|^2 - \|S\|^2 = \|D\|^2 - 2\langle S, D \rangle.$$

Consequently, if D involves at most 4 atom codes and all pairwise inner products between atom codes are bounded by L, then $\|D\|^2 = O(1)$, $\langle S, D\rangle = O(1) + O(kL)$, and

$$\big|\|S'\| - \|S\|\big| = \frac{\big|\|S'\|^2 - \|S\|^2\big|}{\|S'\| + \|S\|} = O\Big(\frac{1 + kL}{\sqrt{k}}\Big).$$

Proof. Expand $\|S - D\|^2$. The consequence follows since $\|S\|, \|S'\| = \Theta(\sqrt{k})$ when kL is bounded away from 1 (shown below). □

The point: the *naive* bound $|\|S'\| - \|S\|| \le \|D\| = O(1)$ loses a factor $\sqrt{k}$, and a margin computed that way carries an error $O(1/\sqrt{k})$ that swamps the true $\Theta(1/k)$ signal. Every margin bound below routes through the identity.

E.4 Random factored codes and their primitives

Definition E.4 (Tied factored sign codes)**.** Draw $u(o) \in \{\pm 1\}^N$ for each object and $v(a) \in \{\pm 1\}^N$ for each attribute, i.i.d. uniform. The *factored code* is the tied pooled code with

$$W(o, a) := \frac{u(o) \odot v(a)}{\sqrt{N}} \qquad (\odot = \text{coordinatewise product}),$$

which is exactly unit-norm. (This is the classical Hadamard binding of vector-symbolic architectures [10, 11].)

All quantities in the margin computations are linear in the following *primitives*:

$$\frac{\langle u(o), u(o')\rangle}{N}, \quad \frac{\langle v(a), v(a')\rangle}{N}, \quad \frac{\langle u(o) \odot u(o'), v(a) \odot v(a')\rangle}{N}, \quad \frac{\langle u(o) \odot u(o'), v(a)\rangle}{N}, \quad \frac{\langle v(a), \mathbf{1}\rangle}{N},$$

for distinct arguments. Each is an average of N i.i.d. ± 1 variables (coordinatewise products of independent Rademacher vectors are Rademacher), and there are at most $4(MK)^2$ of them.

Lemma E.5 (Primitive control)**.** *Fix a failure probability $\eta \in (0,1)$ (reserving δ for the smoothness gap of §G) and set*

$$L^* := \sqrt{\frac{2\log\big(8(MK)^2/\eta\big)}{N}}.$$

With probability at least $1-\eta$ over the codebooks, every *primitive is at most L^* in absolute value simultaneously.*

Proof. Hoeffding: an average of N i.i.d. ±1's exceeds t in absolute value with probability at most $2e^{-Nt^2/2}$. Union bound over at most $4(MK)^2$ primitives: $4(MK)^2 \cdot 2e^{-Nt^2/2} \le \eta$ iff $t \ge L^*$. ☐

Remark E.6. The union bound runs over the $O((MK)^2)$ *primitives*, not over the exponentially many axiom instances—every instance margin is a fixed linear combination of primitives. This is why the dimension requirement below scales as $\log(MK)$ and not $k\log(MK)$.

On the event of Lemma E.5: the factored code has $g = 1$ exactly (tied), and every pairwise inner product between distinct atom codes is a primitive, hence $\ell, \ell_0 \le L^*$. Note also the necessity of tying: if $W_{\mathbf{i}}$ and $W_{\mathbf{t}}$ are drawn *independently*, then g itself is a primitive-sized quantity $O(L^*)$—the same order as the leakage—and every margin collapses. Tying (or training toward alignment) is a real hypothesis, not a convenience.

E.5 The swap margin, in full

Theorem E.7 (Swap margin of the factored code)**.** *Let all primitives be bounded by L (a bound that dominates the ℓ, ℓ_0 of Definition E.2) and suppose $kL \le 1/5$, $k = |s| \ge 2$. For every scene (distinct objects and attributes) and every swap,*

$$\Big| M_{\text{swap}} - \frac{2}{k} \Big| \;\le\; 19\,L.$$

In particular, for $L < 2/(19k)$ the margin is $\Theta(1/k)$ from both sides*: pooled codes do not merely achieve a positive margin, they are pinned at $2/k$. (At the edge of the stated regime $kL \le 1/5$ the interval is vacuous; the typical case $L \approx \sqrt{1/N}$ is far inside it.)*

Proof. Write $W_j := W(o_j, a_j)$, $S := \sum_{j\le k} W_j$, and for the swap on positions $1, 2$:

$$S' = S - D, \qquad D = W(o_1,a_1) + W(o_2,a_2) - W(o_1,a_2) - W(o_2,a_1).$$

Because the code is tied, $\mathbf{i}(I_s) = \mathbf{t}(c(s)) = S/\|S\|$ and $\mathbf{t}(c(\sigma s)) = S'/\|S'\|$, so

$$M_{\text{swap}} = 1 - \frac{\langle S,\, S'\rangle}{\|S\|\,\|S'\|} = 1 - \frac{p-a}{\sqrt{pq}}, \qquad p := \|S\|^2,\;\; q := \|S'\|^2,\;\; a := \langle S,\, D\rangle.$$

Bounds on the pieces. (1) Distinct positions share no object and no attribute, so $\langle W_j,\, W_l\rangle$ ($j \ne l$) is a quadruple primitive; hence $p \in [k - k(k-1)L,\; k + k(k-1)L] \subseteq k[1-kL,\, 1+kL]$, and—since S' is also a sum of k atom codes with distinct objects and attributes—likewise $q \in k[1-kL,\, 1+kL]$. (2) $a = \sum_j \langle W_j,\, D\rangle$. For $j = 1$: the four inner products are 1 (exact match), one quadruple primitive, one attribute primitive, one object primitive, so $\langle W_1,\, D\rangle \in [1-3L,\, 1+3L]$; same for $j = 2$; for $j \ge 3$ all four are primitives: $\langle W_j,\, D\rangle \in [-4L, 4L]$. Hence $a \in [\,2-(4k-2)L,\; 2+(4k-2)L\,] \subseteq [2-4kL,\, 2+4kL]$. (3) $\|D\|^2 = 4 + 2(\text{six signed primitives}) \in [4-12L,\, 4+12L]$. (4) By Lemma E.3, $p - q = 2a - \|D\|^2 \in [-(8k+12)L,\, (8k+12)L]$; for $k \ge 2$, $(8k+12)L \le 14kL$.

Assembling. Split

$$M_{\text{swap}} = 1 - \sqrt{p/q} + \frac{a}{\sqrt{pq}}.$$

Set $x := (p-q)/q$, so $\sqrt{p/q} = \sqrt{1+x}$. From $q \ge k(1-kL) \ge \frac{4}{5}k$ and the two bounds on $|p-q|$ (the cancellation bound $14kL$, and the crude bound $|p-q| \le 2k^2L$ from $p, q \in k[1 \pm kL]$):

$$|x| \;\le\; \frac{14kL}{\frac{4}{5}k} = 17.5\,L, \qquad |x| \;\le\; \frac{2k^2L}{\frac{4}{5}k} = 2.5\,kL \;\le\; \tfrac{1}{2}.$$

For the upper side, concavity gives $\sqrt{1+x} \le 1 + \frac{x}{2} \le 1 + 8.75\,L$. For the lower side use the identity $1 - \sqrt{1+x} = \frac{-x}{1+\sqrt{1+x}}$: with $x \ge -\frac{1}{2}$ we have $\sqrt{1+x} \ge \sqrt{1/2}$, hence

$$1 - \sqrt{1+x} \;\le\; \frac{|x|}{1+\sqrt{1/2}} \;\le\; 0.586\,|x| \;\le\; 10.26\,L, \quad \text{i.e.} \quad \sqrt{p/q} \;\ge\; 1 - 10.26\,L.$$

For the last term, $\sqrt{pq} \in k[1-kL, 1+kL]$, so

$$\frac{a}{\sqrt{pq}} \;\ge\; \frac{2-4kL}{k(1+kL)} \;\ge\; \frac{(2-4kL)(1-kL)}{k} \;\ge\; \frac{2}{k} - 6L, \qquad \frac{a}{\sqrt{pq}} \;\le\; \frac{2+4kL}{k(1-kL)} \;\le\; \frac{2}{k} + 8L,$$

using $kL \le 1/5$ in the last step ($\frac{1}{1-kL} \le 1 + \frac{5}{4}kL$). Combining,

$$\frac{2}{k} - 6L - 8.75L \;\le\; M_{\text{swap}} \;\le\; \frac{2}{k} + 8L + 10.26L,$$

i.e. $|M_{\text{swap}} - 2/k| \le 18.26L \le 19L$. Constants are not optimized. ☐

Remark E.8 (Numerical check). The bound was verified against simulation with real random sign codes: $M{=}K{=}30$, 400 scenes per configuration. At $N = 4096$ the measured mean swap margins are 0.9995, 0.4999, 0.2493 for $k = 2, 4, 8$, against $2/k = 1, 0.5, 0.25$: concentration *at* $2/k$ to three decimals, consistent with the two-sided pinch (typical primitive size is $\approx \sqrt{1/N}$, far below the worst-case L^*).

The same computation with a replace negative ($D = W(o_1, a_1) - W(o_1, a')$, two codes instead of four) gives $M_{\text{repl}} \ge 1/k - 7L$; an add negative gives, exactly at orthogonality and stably under $O(L)$ perturbation,

$$M_{\text{add}} = 1 - \sqrt{\frac{k}{k+1}} - O(L) \;\ge\; \frac{1}{2(k+1)} - O(L).$$

(The exact form $1 - \sqrt{k/(k+1)}$ matters: simulation gives 0.1831, 0.1056, 0.0571 for $k = 2, 4, 8$, matching it to three decimals, whereas the first-order approximation $\frac{1}{2(k+1)}$ is visibly off at $k = 2$.) Drop is symmetric to add. *Swap and replace are proved in full; the add case is outlined — derivative bookkeeping around the orthogonal point is routine and omitted.*

Two structural readings (Figure 10 shows all three margins tracking their rates). First, *all* single-edit margins are $\Theta(1/k)$: the margin budget of a pooled representation is systemic, so (C_γ) is meaningful only for $\gamma \lesssim 1/k$. Second, the swap margin is the *largest* of the family ($2/k$: a swap edits two atoms)—pooled codes are not intrinsically worse at swaps than at replaces; what distinguishes swaps is the text side of §G and the data side of §F.

E.6 Constituent ranking fails for the plain code; augmentation

A finding of the constants pass that corrects our own earlier sketch: the factored code, as defined, *cannot* satisfy (K_k).

Proposition E.9 (The unbinding obstruction)**.** *For the factored code, embed the single-object caption as* $\mathbf{t}(x) = u(x)/\sqrt{N}$. *Then for every scene* s *and every object* x *(present or absent),*

$$\langle \mathbf{i}(I_s), \mathbf{t}(x) \rangle = \frac{1}{\|S\|} \sum_{j \le k} \rho_j, \qquad \rho_j := \frac{\langle u(o_j) \odot v(a_j), u(x) \rangle}{N},$$

and every ρ_j *is a primitive-sized quantity: for* $x = o_j$, $\rho_j = \langle v(a_j), \mathbf{1} \rangle / N$ *(the mean of* $v(a_j)$*), and for* $x \ne o_j$ *a triple primitive. Hence all scores, of present and absent objects alike, lie in* $[-c\sqrt{k}\,L^*, c\sqrt{k}\,L^*]$*: the code offers* no signal *separating* $V(s)$ *from its complement, and* (K_k) *cannot be certified at any fixed margin. In simulation the* (K_k) *event (all present objects outranking all absent ones) occurs at rate* 0.00–0.01.

This is the classical vector-symbolic fact that bound products are dissimilar to their factors—one *unbinds* to query, one does not take inner products with raw factors [10, 11]—resurfacing as an axiom failure. The repair is to pool an object-presence term alongside the bound pair:

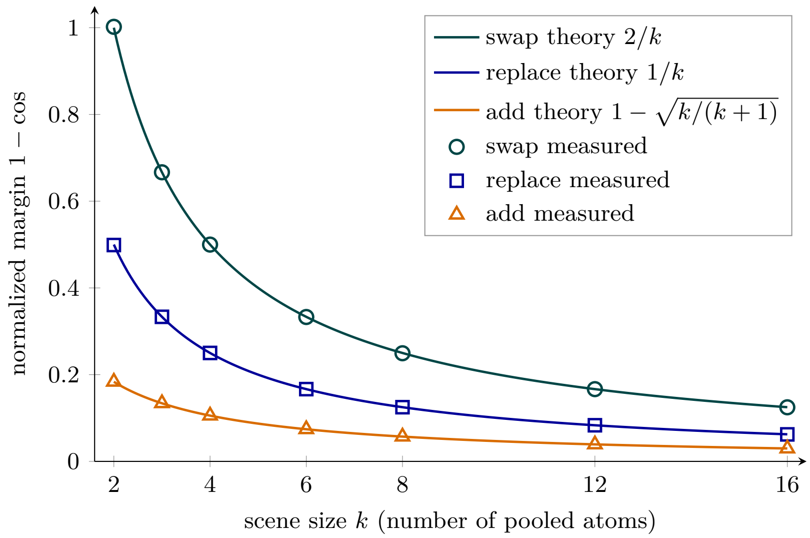


Figure 10: The margin ceiling is $\Theta(1/k)$ (Theorem E.7). Marks: measured normalized margins of the tied factored sign code ($N = 4096$, $M = K = 40$, 300 scenes per size) against swap, replace, and add negatives. Lines: the theoretical $2/k$, $1/k$, and $1 - \sqrt{k/(k+1)}$. Measured values sit on the predictions to plotting accuracy—in particular the *exact* add-margin form $1 - \sqrt{k/(k+1)}$, not the loose $1/(2(k+1))$.

Theorem E.10 (Augmented factored code)**.** *Define* $\widehat{W}(o,a) := \big(u(o) \odot v(a) + u(o)\big)/\sqrt{2N}$ *and pool as before (each* $\widehat{W}$ *is unit-norm up to* $O(L^*)$*, absorbed into the constants). On the event of Lemma E.5 with* $kL^* \le 1/5$*:*

- *(i)* (K_k) *holds with margin* $\Theta(1/\sqrt{k})$*: present objects score* $\frac{1}{\sqrt{2k}}(1 - O(kL^*))$*, absent objects* $O(\sqrt{k}\,L^*)$*.* (Outlined: the computation is the display above plus the new primitive types; simulation pass rate 1.00 in-regime.)
- *(ii) The* attribute-edit *margins are exactly halved: the presence part* $u(o)$ *cancels in any difference that preserves the object multiset (swap, attribute-replace), while the* $\sqrt{2}$ *normalization dilutes the surviving bound part. Add/drop and* object-*replace margins are* unchanged*—their differences carry a presence term. Quantitatively,* $|M_{\text{swap}} - 1/k| \le 27L$*.* (Proved by the route of Theorem E.7, adjusted for the augmented code's weaker primitives ($p, q \in k[1 \pm 2kL]$, forcing the cruder $1 - \sqrt{1+x} \le |x|$ step; constant unoptimized). Simulation: augmented swap margin 0.1243 vs. half 0.1247 at $k = 8$; augmented object-replace and add margins match the *plain* values to 3 decimals.)
- *(iii) A mixing weight* $(u \odot v + \alpha u)$ *trades the two margins continuously; no single pooled vector maximizes both.* (Outlined: the (K_k) margin is increasing and the attribute-edit margins decreasing in α.)

Part (iii) deserves emphasis: *within* the pooled class, the retrieval axiom (K_k) and the binding axioms compete for the same normalization budget. This is a miniature, class-internal echo of the global tension that §G proves for arbitrary encoders.

E.7 Dimension rates and the leakage dichotomy

Corollary E.11 (Binding is cheap if the code is right)**.** *Rates are keyed to the instance family they certify: the uniform margin over the whole hard core is set by its weakest member—add/drop,* $\approx \frac{1}{2(k+1)}$ *(Remark B.4(2))—so no bound of the form* $\gamma = c/k$ *with large* c *can cover the full family. For the plain factored code: the* swap *instances hold at* $\gamma = c/k$,

$c < 2$, *once* $L^* \le (2-c)/(19k)$, *i.e.*

$$N \ \ge \ \frac{722\, k^2 \ \log\bigl(8(MK)^2/\eta\bigr)}{(2-c)^2};$$

the replace *instances at* $\gamma = c/k$, $c < 1$, *once* $L^* \le (1-c)/(7k)$ *(*$N \ge 98\, k^2 \log(8(MK)^2/\eta)/(1-c)^2$*); and a* uniform *margin* $\gamma = c/(2(k+1))$, $c < 1$, *over the whole hard core at* $N = O(k^2 \log MK)$ *(the add/drop constant is Outlined). For the augmented code the swap instances hold at* $\gamma = c/k$, $c < 1$, *once* $L^* \le (1-c)/(27k)$, *i.e.* $N \ge 1458\, k^2 \log(8(MK)^2/\eta)/(1-c)^2$, *with* (K_k) *joining at margin* $\Theta(1/\sqrt{k})$*. In every case: logarithmic in the number of concepts, quadratic in scene size. (Constants from Theorems E.7 and E.10(ii); none optimized.)*

The encoder-class landscape is now a three-way dichotomy organized by leakage: $\ell \approx 0$ (factored codes: all axioms satisfiable at $N = O(k^2 \log MK)$); $\ell = \Theta(g)$ with pairing symmetry (additive codes: margin exactly 0, Theorem D.2); ℓ uncontrolled (generic learned codes: no guarantee, and Paper 1 measures trained MLP composition heads at $\ell \approx +0.17$ with below-chance extrapolation to unseen pairings as the consequence). *It is the code, not the pooling*—now with a rate.

Remark E.12 (Pooled atom codes have exactly the required invariance). In the language of Definition B.5: a pooled code over *bound pairs* is a set sum, hence invariant under the order variants within $\mathcal{C}(s)$ (every order variant of a scene receives the identical embedding—zero intra-class spread) while separating $\mathcal{C}(\sigma s)$ at margin $\Theta(1/k)$ (Theorem E.7), because the swap changes *which pairs* are summed. Pooling over *words* (the additive code) is invariant under both, fatally (Remark B.6). Pooling is thus compatible with binding precisely when the pooled units are the bound pairs—a second, sharper sense of "it is the code, not the pooling."

F Obstruction III: the training distribution

The placements of §§C–E answer "could vectors exist"; real encoders are *selected by an objective on data*. This section states, at the level of precision they currently have, the two data-side laws established in Paper 1, because the measurements of §§H–I constantly refer to them. Both are ported from Paper 1: formal statements with proof outlines and quantitative empirical support, not theorems of the present paper.

Lemma F.1 (What contrastive training optimizes; outline)**.** *Population InfoNCE with matching pairs* $(I, c) \sim p$ *and negatives* $c' \sim Q$ *is minimized (over unconstrained score functions) at* $s^*(I, c) = \log \frac{p(c|I)}{Q(c)} + h(I)$ *for an arbitrary per-image function* h *[19]. Consequently the objective carries signal for separating* $c(s)$ *from* $c(\sigma s)$ *only insofar as the pair* (p, Q) *distinguishes them on observed data.*

Proposition F.2 (Throttle and coverage; ported)**.** *Let* π *denote the rate at which the negative process presents the* swap *of the positive caption.*

(i) ***(Throttle.)*** *The* trainable *swap margin—the margin attainable by gradient training on* D*—scales linearly in* π *for small* π*. With* in-batch *negatives (the other captions of the same training batch) over a caption corpus of size* $|C|$, $\pi = O(1/|C|)$*: asymptotically no swap signal, and the selected optimum is a bag-of-words-equivalent solution violating every swap instance of* (C_γ)*.* Measured *(Paper 1, real-COCO* π*-sweep): swap accuracy rises* $0.62 \to 0.70$*, concavely, as* $\pi : 0 \to 1$*, with retrieval metrics flat; a loss-weight dial* λ *is near-equivalent to* π*.*

(ii) ***(Coverage.)*** *For atom pairs never co-observed in training, the loss is flat in their relative placement: held-out binding is decided by the encoder class alone. Factored-structure codes generalize along an occupancy law* $\mathrm{acc}(\rho) \approx \frac{1}{2} + \frac{1}{2}(1-e^{-\rho K})(1-e^{-\rho M})$ *in the pair-density* ρ *(the expected number of training exposures per object–attribute pair), with threshold at* $\rho \approx 2$–3*; additive codes sit at chance (Theorem D.2); generic MLP codes extrapolate* below *chance via the leakage mechanism of Definition E.2.* Measured *(Paper 1 five-arm code ablation; external refit to two published training curves).*

The reading that matters downstream: the axioms of Definition B.8, pushed through training, become *conditions on the data distribution*—a positive swap rate π on the contrasted families, and either pair coverage or a factored-class code. Neither condition holds for standard web-scale contrastive training, which is Paper 1's explanation of the status quo. The present paper asks what remains *after* those conditions are met; the answer is the frontier. (Appendix K upgrades part (i) of the proposition to proved, architecture-free theorems.)

G The smoothness–binding frontier

G.1 The two-jobs tension

An embedding space in deployment serves two masters:

- **Semantic similarity.** Paraphrases and near-variants should embed close together—this is what retrieval, similarity search, and deduplication consume, and what similarity benchmarks reward.
- **Binding.** $c(s)$ and $c(\sigma s)$ should embed far apart—this is what swap benchmarks reward.

The tension is that a swap is a *minimal* semantic edit: same words, same objects, same attributes, different wiring. The SugarCrepe++ literature [20] documents the dissociation empirically (strong hard-negative performance need not transfer to paraphrase consistency, and several compositionality fine-tunes degrade on it), but to our knowledge no quantitative statement existed. The frontier theorem makes the trade-off exact.

G.2 Anchors, smoothness, and the guard against tautology

Definition G.1 (Pairing-neutral anchor; δ-smoothness)**.** Fix a scene s. A *pairing-neutral anchor* is any unit vector $\bar{t}_s$ that depends only on the *symbol multiset* of s (Definition B.1)—not on the pairing. (Examples: the embedding of the bag-of-atoms caption "a photo of a car and a dog, red and blue"; the mean embedding of a paraphrase family generated without pairing information.) The caption map is *δ-smooth at s with respect to $\bar{t}_s$* if both pairings lie in the spherical cap of cosine-radius $1-\delta$ about it:

$$\langle \mathbf{t}(c(s)), \bar{t}_s\rangle \;\geq\; 1-\delta \qquad \text{and} \qquad \langle \mathbf{t}(c(\sigma s)), \bar{t}_s\rangle \;\geq\; 1-\delta.$$

Why the anchor must be pairing-neutral—and why the definition would otherwise be vacuous—is exposed by the following computation, which we will reuse twice.

Lemma G.2 (Midpoint identity)**.** *Let $t_1, t_2 \in \mathbb{S}^{N-1}$, $d := \|t_1 - t_2\|$, and let $\bar{m} := (t_1 + t_2)/\|t_1 + t_2\|$ be their normalized midpoint. Then both t_i satisfy $\langle t_i, \bar{m}\rangle = 1 - \delta_{\mathrm{mid}}$ with*

$$d \;=\; 2\sqrt{2\delta_{\mathrm{mid}}} \cdot \sqrt{1-\delta_{\mathrm{mid}}/2}.$$

Proof. $\langle t_i, \bar{m}\rangle = \frac{1+\langle t_1, t_2\rangle}{\|t_1+t_2\|} = \frac{\|t_1+t_2\|}{2}$, so $2\delta_{\mathrm{mid}} = 2 - \|t_1+t_2\|$. Then $d^2 = 2 - 2\langle t_1, t_2\rangle = 4 - \|t_1+t_2\|^2 = (2 - \|t_1+t_2\|)(2+\|t_1+t_2\|) = 2\delta_{\mathrm{mid}}(4-2\delta_{\mathrm{mid}})$. □

So *against its own midpoint, every pair sits exactly on the curve $d = 2\sqrt{2\delta}$ (to leading order).* If the anchor were allowed to depend on the pair, the theorem below would be an identity with zero content. The force of Definition G.1 is the quantifier: the bound holds for *every* pairing-neutral anchor, so exhibiting any *one* independently defined anchor near both variants yields a genuine, non-circular ceiling. Measurement protocols must (and ours do) use an anchor constructed without reference to the pairing.

G.3 The theorem

Theorem G.3 (Smoothness–binding frontier)**.** *Let s be a scene and suppose the caption map is δ-smooth at s with respect to some pairing-neutral anchor $\bar{t}_s$.*

(i) ***(Upper bound.)*** *For* every *image map, every dimension N, and every training procedure,*

$$M(s) \;\le\; \|\mathbf{t}(c(s)) - \mathbf{t}(c(\sigma s))\| \;\le\; 2\sqrt{2\delta}.$$

(ii) ***(Tightness.)*** *For every $\delta \in (0, 1)$ there exist unit vectors t_1, t_2, a pairing-neutral anchor, and a unit image vector achieving*

$$M(s) \;=\; 2\sqrt{2\delta}\cdot\sqrt{1-\delta/2} \;=\; 2\sqrt{2\delta}\,\bigl(1 - O(\delta)\bigr),$$

so the rate $\sqrt{\delta}$ and the constant $2\sqrt{2}$ are exact.

Proof. (i) First inequality: Cauchy–Schwarz with $\|\mathbf{i}(I_s)\| = 1$. Second: for any unit x with $\langle x, \bar{t}_s\rangle \ge 1 - \delta$,

$$\|x - \bar{t}_s\|^2 = 2 - 2\langle x, \bar{t}_s\rangle \le 2\delta,$$

so both variants lie within $\sqrt{2\delta}$ of the anchor, and the triangle inequality through $\bar{t}_s$ gives $\|\mathbf{t}(c(s)) - \mathbf{t}(c(\sigma s))\| \le 2\sqrt{2\delta}$. (The exact diameter of the cap is $2\sqrt{2\delta - \delta^2}$; we quote the clean form.)

(ii) Work in a 2-plane spanned by orthonormal $\bar{t}, w$; put $t_{1,2} = \cos\theta\,\bar{t} \pm \sin\theta\, w$ with $\cos\theta = 1 - \delta$. Declare $\bar{t}$ the anchor (it is a fixed vector; in the full-system construction of Proposition C.3(ii) it can be realized as the embedding of the atom-list caption, which carries no pairing information). Both variants satisfy the smoothness condition with equality (Lemma G.2 is this configuration seen from the midpoint; Figure 11 draws the cap geometry and the induced Pareto region). Take the image vector $\mathbf{i} := w$. Then

$$M(s) = \langle w, t_1 - t_2\rangle = 2\sin\theta = 2\sqrt{1-(1-\delta)^2} = 2\sqrt{2\delta - \delta^2}. \qquad \square$$

Corollary G.4 (Per-item ceiling; anchor-free)**.** *For every scene and every image map,*

$$\bigl|M(s)\bigr| \;\le\; d(s) := \|\mathbf{t}(c(s)) - \mathbf{t}(c(\sigma s))\|.$$

The corollary is trivial (it is the Cauchy–Schwarz half alone) but operationally central: $d(s)$ is computable *from the text tower alone*, yet it caps the margin of the full image–text system on that item. §I tests precisely this quantity on a real benchmark.

Corollary G.5 (Class-level frontier)**.** *Suppose the caption map is δ-smooth* as a class *at s: every realization in $\mathcal{C}(s) \cup \mathcal{C}(\sigma s)$ lies within cosine $1-\delta$ of a pairing-neutral anchor $\bar{t}_s$. Then for every image realization $I \in \mathcal{I}(s)$ and every cross-class pair $(c^+, c') \in \mathcal{C}(s) \times \mathcal{C}(\sigma s)$,*

$$\langle \mathbf{i}(I),\, \mathbf{t}(c^+) - \mathbf{t}(c')\rangle \;\le\; 2\sqrt{2\delta}.$$

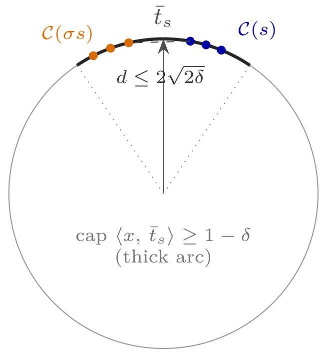
(Proof: verbatim Theorem G.3(i) applied to the pair.) *This is the formal version of the two-jobs tension of §G.1: the similarity job asks each class to be* compact *(small spread around its meaning), the binding job asks the two classes to be* separated*—and both classes share the lexical neighborhood of the same anchor, so compactness of the union caps the separation. The anchor-as-paraphrase-centroid used in the measurements of §H is the natural instance, not a convenience.*

Remark G.6 (Scope: what the theorem does and does not say). (1) The bound is *per scene* and *anchor-relative*; its empirical content on a model is only as strong as the independence of the anchor used to measure δ. (2) Being far inside the frontier is not a virtue and being on it is not a vice: the theorem constrains the *jointly achievable* pairs (δ, M), and where a model should sit depends on its deployment mix of the two jobs. (3) Nothing is claimed about *which* models approach the bound—that is an empirical question, answered in §H: none do, yet. (4) The exact-collapse claim of Kang et al. is recovered as the endpoint $\delta = 0$ (anchor equal to both variants forces $M(s) \le 0$); the frontier is its finite-δ, two-sided completion.

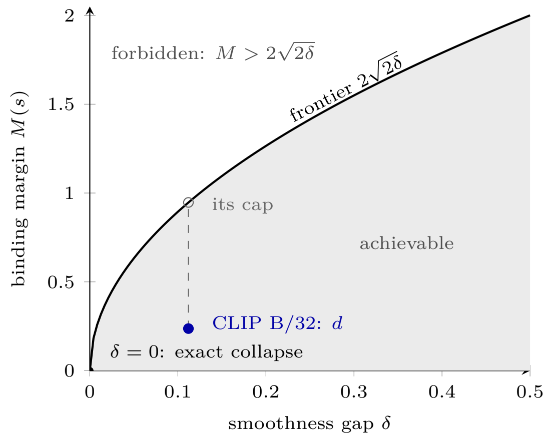
G.4 The diagnostic

Theorem G.3 induces a text-only measurement. For a model and a scene family:

$$d \;=\; \|\mathbf{t}(c(s)) - \mathbf{t}(c(\sigma s))\|, \qquad \delta \;=\; 1 - \min\bigl(\langle \mathbf{t}(c(s)),\, \bar{t}_s\rangle,\, \langle \mathbf{t}(c(\sigma s)),\, \bar{t}_s\rangle\bigr),$$



(a)



(b)

Figure 11: The smoothness–binding frontier. **(a)** Proof geometry of Theorem G.3(i) (drawn as a great-circle section): both caption classes lie in the spherical cap of cosine-radius $1-\delta$ about the pairing-neutral anchor $\bar{t}_s$; any two points of the cap are at distance $\le 2\sqrt{2\delta}$, and by Cauchy–Schwarz no unit image vector can extract a larger margin from them (Corollary G.5 is exactly this picture). **(b)** The induced Pareto region: pairs (δ, M) below the curve are achievable (Theorem G.3(ii) saturates it), pairs above are impossible for *any* encoder, dimension, or training. Kang et al.'s exact collapse is the $\delta = 0$ endpoint. The measured CLIP B/32 point sits at 25% of its cap—far inside; the dashed segment is the unused smoothness budget (§H). The open circle sits on the curve at $2\sqrt{2 \cdot 0.112} = 0.947$; the table of §H reports the mean of per-quartet caps, 0.936—a Jensen gap.

$$\text{cap} \;=\; 2\sqrt{2\delta}, \qquad \textit{frontier usage} \;=\; \frac{d}{\text{cap}} \in [0,1].$$

Usage ≈ 1: the model spends its entire smoothness budget on separating the pairings — binding is *smoothness-limited*, and further binding gains must be paid for in paraphrase geometry. Usage $\ll 1$: the ceiling is far away, and binding is limited by something else—by Paper 1's results, by the code and the data. Which regime deployed models occupy is an empirical question.

H Measurement I: the frontier across the model zoo

H.1 Protocol

Scene family: a *quartet* is a choice of two objects and two attributes, determining a 2-scene and its swap; all $\binom{10}{2}\binom{6}{2} = 45 \cdot 15 = 675$ quartets from 10 objects $\times$ 6 color attributes (2-object scenes; captions "a $\{a_1\}$ $\{o_1\}$ and a $\{a_2\}$ $\{o_2\}$" and the attribute swap). Anchor: the embedding of the pairing-neutral atom-list caption "a photo of a $\{o_1\}$ and a $\{o_2\}$, $\{a_1\}$ and $\{a_2\}$"—constructed from the symbol multiset only. Every model receives the identical quartet grid and anchor template; text towers are ℓ_2-normalized (the geometry of Definition B.7). In Table 1, 'ft' abbreviates fine-tune, 'LoRA-4' a rank-4 low-rank adapter, CC3M the Conceptual Captions 3M corpus, and the 'max' column is the model's largest per-quartet usage. Aggregate usage is $\overline{d}/\overline{\text{cap}}$ with a 2000-resample bootstrap 95% CI. Implementation checks for NegCLIP and the LoRA models (SVLC [25]; DAC [26]) are in §H.4.

H.2 Results

The full grid results appear as Table 1 in the main text; Figure 3 plots them.

H.3 Findings

1. **A universal band, far inside the frontier** (Figure 3). Thirteen base models spanning three training objectives (InfoNCE softmax, sigmoid, and ALIGN's noisy InfoNCE), five pretraining corpora, two text architectures (GPT-style Transformer, BERT), and a 10× scale range all land at usage 0.25–0.32. *No base model is within* 3× *of the ceiling* (NegCLIP, at usage 0.349, reaches 2.9×). Binding failure in the current zoo is therefore *not* smoothness-limited—consistent with, and required by, Paper 1's account (code and data are the binding constraints today). The frontier is the *terminal* obstruction: the one that will remain once those are fixed.
2. **Scale moves models along the frontier, not toward it.** Larger towers increase d and cap together (CLIP B/32 → L/14: d 0.237 → 0.293, cap 0.936 → 1.131), leaving usage nearly fixed.
3. **Real caption geometry traces the predicted direction.** Within every model, quartets with larger d sit at larger δ (correlation +0.10 to +0.82, positive in all 18 models)—directionally consistent with the frontier's d–δ coupling (no exponent was fitted).
4. **Mechanism separation in the fine-tuned family.** Usage can rise two ways: raising d (more binding signal) or shrinking the cap (smoother anchors, lower ceiling). Only NegCLIP—a *full* fine-tune with caption-level swap negatives, i.e. Paper 1's throttle applied in the wild—raises d (0.237 → 0.320 against its architecture-matched baseline at flat δ), climbing *toward* the frontier (0.253 → 0.349, disjoint CIs). Three of the four LoRA rank-4 compositional fine-tunes (DAC-SAM, both SVLC) leave d at or below baseline and move usage through δ; DAC-LLM raises d modestly (0.262) at flat δ. *Caveat*: NegCLIP differs from the LoRA models in tuning depth, data, and negative type simultaneously; the adapter-capacity explanation is a hypothesis, not a conclusion.

H.4 Measurement rigor

Three checks guard the comparisons. (i) *Architecture matching*: NegCLIP fine-tunes OpenAI's QuickGELU ViT-B/32; loading it into a GELU architecture silently corrupts outputs, so both it and its baseline use the QuickGELU graph. (ii) *Framework neutrality*: the OpenAI baseline evaluated through the open_clip stack reproduces the HuggingFace-stack number to four decimals (usage 0.2534 in both). (iii) *LoRA correctness gate*: DAC and SVLC ship as rank-4 LoRA adapters whose fused-QKV forward cannot be reproduced by naive weight merging; we run the authors' own fork, and verify that *zeroing the adapters reproduces the baseline* (usage 0.255 on the 150-quartet gate subset, equal to the fork's own baseline there; the full-grid baseline is 0.253)—so the reported deltas are attributable to the adapters and not to the loading path.

I Measurement II: the ceiling on a real benchmark

Corollary G.4 says $|M(s)| \leq d(s)$ *per item*, with $d(s)$ computable from text alone. If the theory describes reality, d should behave like a ceiling on real benchmark decisions: not a predictor of success, but a bound—low-d items *must* sit near chance (their margin is squeezed to noise), high-d items are *permitted* to be solved. We test this on SugarCrepe [3].

I.1 Protocol

All five SWAP/REPLACE subsets of SugarCrepe (add omitted; Remark B.4): 4757 items, each an image with a positive caption and one hard negative; 1542 distinct COCO val2017 images. Fourteen of the eighteen models (four skipped for CPU compute: OpenCLIP L/14 and H/14, SigLIP-large, and CLIP L/14-336, whose text tower duplicates CLIP L/14; the scale axis is covered by CLIP L/14). For each model and item we record the actual decision quantity and its text-side ceiling:

$$\text{margin} = \langle \mathbf{i}(I), \mathbf{t}(c_{\text{pos}}) \rangle - \langle \mathbf{i}(I), \mathbf{t}(c_{\text{neg}}) \rangle, \qquad d = \|\mathbf{t}(c_{\text{pos}}) - \mathbf{t}(c_{\text{neg}})\|,$$

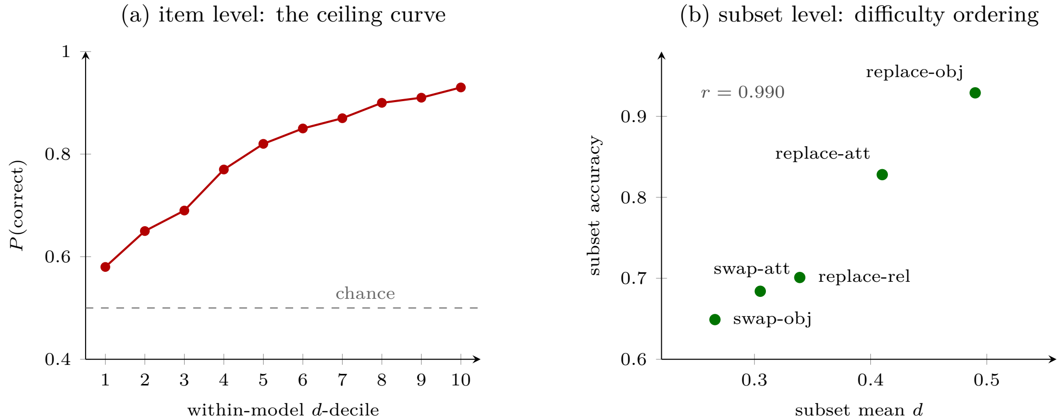


Figure 12: The per-item ceiling (Corollary G.4) on SugarCrepe, 14 models, 66,598 decisions. **(a)** Accuracy by within-model d-decile, pooled: monotone from near-chance to 0.93; 73% of all errors lie below the median d. The low end is the theorem (small $d \Rightarrow$ margin within noise); the high end is permitted-and-observed, not implied. **(b)** Pooled subset accuracy against subset mean d: the benchmark's canonical difficulty ordering—swaps hardest—is the statement that same-vocabulary negatives have small ceilings.

correct iff margin > 0. Sanity anchor: CLIP B/32 reproduces published SugarCrepe accuracy (0.775 overall; swap-object 0.61). In total $14 \times 4757 = 66{,}598$ decisions.

I.2 Three levels of validation

Subset level: d explains the benchmark's difficulty ordering ($r = 0.990$; Figure 12b). Pooling across models:

SugarCrepe subset	mean d	accuracy
swap-object	0.266	0.649
swap-attribute	0.305	0.684
replace-relation	0.339	0.701
replace-attribute	0.410	0.828
replace-object	0.490	0.929

Pearson correlation of subset mean-d with subset accuracy: 0.990. The reading is structural: swap negatives use the *identical vocabulary* as the positive (Definition B.3), so their d is intrinsically small; replace negatives change a word, so their d is large. *The canonical observation that swaps are the hard subsets of SugarCrepe is, to within $r = 0.99$, the statement that swaps are the small-ceiling subsets.*

Item level: a monotone ceiling curve (Figure 12a). Within each model, rank items by d into deciles; pool the accuracy of each decile across models (66,598 decisions):

$$0.58,\ 0.65,\ 0.69,\ 0.77,\ 0.82,\ 0.85,\ 0.87,\ 0.90,\ 0.91,\ 0.93 \qquad (\text{deciles } 1 \to 10).$$

Monotone from near-chance to near-ceiling, +35 points. Errors localize accordingly: 73% of all incorrect decisions occur on below-median-d items. We stress the logically correct, one-sided reading: the low end is the theorem *plus a noise model* (small d caps $|M|$ by Corollary G.4; near-chance additionally assumes that sub-ceiling margins are sign-symmetric—an empirical regularity, not part of the corollary) (small $d \Rightarrow$ margin within noise $\Rightarrow$ near-chance); the high end is *permitted-and- observed*, not implied—d is necessary, not sufficient.

Model level: the ceiling logic, not naive correlation. Correlating model-level predictors with model accuracy ($n = 14$):

Predictor	Pearson r (overall acc)	Pearson r (swap acc)
usage d/cap (frontier position)	0.45	**0.59**
d on the synthetic probe (§H)	0.40	0.26
raw benchmark mean-d	0.07	−0.15
δ (smoothness)	0.09	−0.16

- *Raw cross-model d predicts nothing*—models differ in global spread (CLIP L/14 spreads all captions more without binding better). Normalizing by the cap removes exactly that scale, and *usage* becomes the best text-only cross-model signal for swap accuracy.
- *δ predicts nothing—and must not.* Every model in §H is far inside the frontier, so its smoothness level is not its binding constraint. The null is the *regime test* passing: were a model on the frontier, δ would become the predictor.
- Case studies make necessity-vs-sufficiency concrete. **NegCLIP** raises the ceiling and converts it into accuracy (swap 0.61 → 0.76). **DAC** binds well (0.850/0.868 overall) at (DAC-SAM) or barely above (DAC-LLM) the baseline ceiling—better exploitation of the existing text separation by the image side, which Corollary G.4 permits and text-side d cannot see. **SVLC-LLM+RB**, the model whose d fell *furthest* below baseline (0.202), records the lowest overall accuracy (0.745): the ceiling coming down is visible in the benchmark.

I.3 Caveats

The quartet probe is synthetic 2-object color–object binding; naturalistic multi-relation captions are future work. The anchor is one template; an anchor-robustness appendix (a second independent anchor on several models) is planned before publication. All results are text-side or single-benchmark; no claim is made that d ranks models on unrelated capabilities.

J Synthesis, related work, open problems

J.1 The obstruction map: three contingent gates, one terminal wall

Obstruction	Hypothesis added	Result	Strength	Status
Geometric	none (free placements)	satisfiable iff $k \le \lfloor N/2 \rfloor$	—	Cited, §C
Structural	pairing-blind code	margin $\equiv 0$, exact	absolute	Proved, §D
Structural	pooled code, leakage L	margins $= \Theta(1/k)$, pinned	quantitative	Proved, §E
Statistical	trained on (p, Q)	swap signal $\propto \pi$; coverage law	quantitative	from [6], §F
Semantic	δ-smooth anchors	$M(s) \le 2\sqrt{2\delta}$, tight	terminal	Proved, §G

The first three obstructions are *contingent* (Figure 13): they can be engineered away (choose a factored code, raise π, cover the pairs). The frontier cannot: it binds *any* encoder, *any* dimension, *any* training, as long as the space is asked to keep near-paraphrases close. That is the sense in which it is terminal—and the measurements say no current model is near it, which returns the near-term agenda to the contingent obstructions; the frontier becomes the binding constraint only after those are repaired.

J.2 Relation to Paper 1

Paper 1 ("It is the code, not the pooling") establishes the contingent story quantitatively: the five-arm code ablation and the partial-match-leakage mechanism (here §E's measured counterpart), and the π-throttle with its dose–response on real data (here Proposition F.2). This paper supplies the principled complement: the axiom system those experiments implicitly assume, the theorems that organize the code classes, and the terminal bound that none of the engineering can move. There is no claim overlap: Paper 1 never states the frontier; this paper claims none of Paper 1's measurements as new.

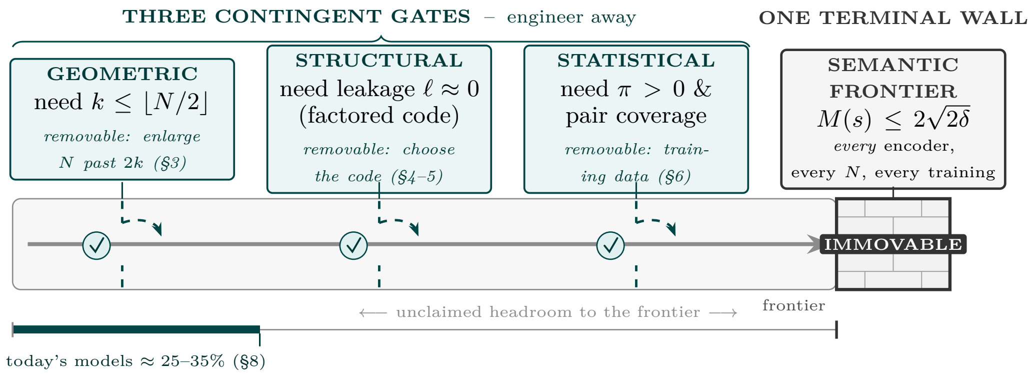


Figure 13: The three-obstruction map (Section 10). Binding faces four obstructions. Three are *contingent* and can be engineered away—enlarge N past $2k$ (geometry, Prop. C.3), choose a low-leakage factored code (structure, §E), and supply swap negatives with pair coverage (data, §F). The fourth, the smoothness frontier $M(s) \leq 2\sqrt{2\delta}$ (Thm G.3), binds *every* encoder, dimension, and training: it is terminal. The measurements (§H) place today's models far short of it, which is why the contingent obstructions govern the near term and the frontier is the limit that remains.

J.3 Related work and differentiation

Three recent lines prove limits on dual encoders by *other* mechanisms, and one documents our tension empirically:

- **Dimension / capacity.** LIMIT [21] bounds which retrieval patterns fit in dimension N via thresholdable-rank arguments (sign-rank in their appendix); [8] pins the minimal dimension for top-k retrieval (the mathematics of §C). Orthogonal axis: those bounds vanish as N grows; the frontier does not involve N at all.
- **Resolution.** "Bound by semanticity" [22] shows identification–generalization limits from finite semantic resolution, assuming non-compositional representations; no swap/margin content, no converse.
- **Optima structure.** Chen et al. [23] prove existence of swap-insensitive contrastive optima; existence only, no quantitative bound, no measurement.
- **Empirics.** SugarCrepe++ [20] documents the paraphrase/hard-negative dissociation that Theorem G.3 formalizes; no bound.

Ours is, to our knowledge, the only *tight two-sided* trade-off among these, and the only one shipped with a measured frontier position for the deployed zoo. The upper bound's proof is elementary—we present that as a feature, not a weakness: the content is in the formulation (the anchor quantifier that evades tautology), the converse, and the measurements.

J.4 Open problems

1. **Margin-robust dimension.** (K_k) with margin γ changes §C's answer from $2k$ to (in related settings) $\Theta(k \log(M/k))$ [24]; the exact constant for the constituent-ranking family (K_k) appears open.
2. **Smoothness vs. neighborliness.** Kang's optional similarity condition (attribute variants cluster near their object) plausibly conflicts with the neighborliness that (K_k) requires at some (M, K, N)—a candidate *second* impossibility theorem with real content.
3. **Breaking the NegCLIP confound.** A LoRA-rank-4 model trained on COCO with swap negatives would isolate which of NegCLIP's three differences moves d.
4. **Naturalistic frontier.** Extending the diagnostic beyond 2-object color binding to relations and counting; anchors for multi-relation scenes are not canonical.

5. **Depth.** The concept system is flat: a scene is a *set* of atoms, and k counts objects, not nesting. The homomorphism form (H) of §E.1 is recursive, and recursion is what makes language productive. Whether the $\Theta(1/k)$ margin of Theorem E.7 degrades gracefully or catastrophically under nesting was open when this program began, and is the natural formal home for the empirical observation that grammar-structured composers help precisely on deeper-than-trained inputs. (Resolved in Appendix L: the margin law $2\,b^{-D}$ is proved — Theorem L.4 — and measured across three orders of magnitude.)
6. **On-frontier models.** No current model is smoothness-limited. A model trained with aggressive π and a factored-structure head might be the first to reach the frontier—at which point δ, not code or data, becomes its binding constraint, and the trade-off of Theorem G.3 becomes an engineering decision.

K The throttle theorems: what the objective pays for binding

Proposition F.2(i) reported, as a ported empirical law, that the trainable swap margin scales with the swap-negative rate π. This section proves the underlying statements. The guiding structural fact — easy to miss and essential for honesty — is that the population InfoNCE optimum with universal capacity *does bind* (Lemma F.1: the optimum is a pointwise-mutual-information scorer, which separates swaps). So the rigorous throttle is *not* "optima do not bind"; it is that **the objective's preference for binding over blindness is a sliver whose width is the swap-contrast rate**, so that anything else — capacity limits, regularization, early stopping, leakage-prone parameterizations — outweighs the objective inside that sliver. Figure 14 draws the landscape.

K.1 Setup

Scenes $s \sim \mathcal{D}$ (finite support, the concept system of §B); caption $c(s)$; image $I_s \sim p(\cdot \mid s)$. A *contrastive sample* is a positive $(I_s, c(s))$ together with a negative caption c^-: with probability π the *explicit swap* $c(\sigma s)$; otherwise a *marginal negative* $c(s')$, $s' \sim \mathcal{D}$ independent. The *implicit collision rate* is $\pi_{\text{impl}} := \Pr_{s,s'}[[c(s')] = [c(s)]]$ — the chance a marginal negative lands in the positive's symbol class (Definition B.1); for web-scale corpora this is vanishing, and in-batch training has $\pi = 0$.

The loss is the one-negative logistic contrastive loss (binary InfoNCE),

$$L(S) \;=\; \mathbb{E}\,\ell\big(S(I_s, c(s)) - S(I_s, c^-)\big), \qquad \ell(m) = \log(1 + e^{-m}),$$

over *arbitrary measurable scorers* S — no architecture assumption; every bound below therefore applies to dual encoders a fortiori. We use: ℓ convex, decreasing, 1-Lipschitz, $\ell(0) = \log 2$, $0 \le \ell(m) \le e^{-m}$ for $m \ge 0$, and the exact identity

$$\ell(-m) - \ell(m) = m. \tag{2}$$

Two comparators are attached to any scorer S:

- the **class-blind average** $\bar{S}(I, c) := \sum_{c' \in [c]} w_{[c]}(c')\, S(I, c')$ (w = the $\mathcal{D}$-conditional weights of the class) — a pairing-blind scorer;
- the **reflection** $(\rho S)(I, c) := S(I, \operatorname{swap}(c))$ — the scorer with its binding *exactly reversed* (on classes with a unique swap partner).

Regularity parameters: Δ-*separation* — for a.e. (s, I_s) and every negative-support c' with $[c'] \neq [c(s)]$, $S(I_s, c(s)) - S(I_s, c') \ge \Delta$ (cross-class contrasts are easy); *within-class spread* R — $|S(I, c) - S(I, c')| \le R$ for $c' \in [c]$; and the matched swap margin $M_S(s) := S(I_s, c(s)) - S(I_s, c(\sigma s)) \le R$.

K.2 The theorems

Theorem K.1 (Blindness is nearly optimal)**.** *For any scorer S with Δ-separation and spread R,*

$$L(\bar{S}) - L(S) \;\le\; (\pi + \pi_{\text{impl}}) \cdot \log 2 \;+\; e^{-(\Delta - 2R)}.$$

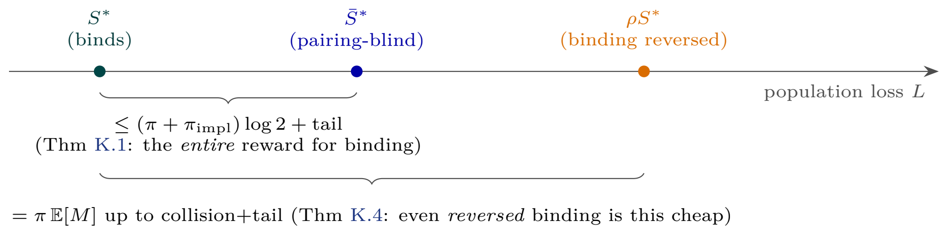


Figure 14: The throttle, schematically. With universal capacity the optimum S^* binds — but its pairing-blind average and even its exact binding reversal sit within a loss sliver whose width is set by the swap-contrast rate $(\pi + \pi_{\text{impl}})$. Training pressures larger than the sliver, not the objective, decide the binding behaviour of the selected solution.

Proof. Condition on the negative type. *Matched contrasts* (c^- in the positive's class: explicit swaps w.p. π, implicit collisions w.p. $\le \pi_{\text{impl}}$): $\bar{S}$ scores positive and negative identically, so its loss is exactly $\ell(0) = \log 2$, while S's is ≥ 0; excess $\le \log 2$. *Unmatched contrasts*: $\bar{S}$ moves each argument by at most R, so its margin is $\ge \Delta - 2R$ and its loss $\le \ell(\Delta - 2R) \le e^{-(\Delta-2R)}$; excess at most that. Weight the cases. □

Corollary K.2 (The throttle)**.** *For any population-optimal S^* with the stated regularity, the pairing-blind $\bar{S}^*$ is ε-optimal with $\varepsilon \le (\pi + \pi_{\text{impl}}) \log 2 + e^{-(\Delta-2R)}$:* the entire objective value of binding — everything beyond bag-of-words — is at most $(\pi + \pi_{\text{impl}}) \log 2$ nats plus an exponential tail. *No training guarantee of the form "ε-optimality" can imply binding unless ε is below this threshold; at web scale the threshold collapses.*

Corollary K.3 (λ–π equivalence)**.** *Up-weighting matched terms by λ multiplies the matched case of the proof exactly as raising π does: the sliver width is $(\lambda\pi + \pi_{\text{impl}})$-scaled. The empirically measured near-equivalence of the loss-weight dial and the rate dial is a one-line consequence.*

Theorem K.4 (Anti-binding costs exactly $\pi\, \mathbb{E}[M]$)**.** *For any scorer S with the regularity above,*

$$\big| L(\rho S) - L(S) - \pi\, \mathbb{E}_s[M_S(s)] \big| \;\le\; \pi_{\text{impl}}\, R \;+\; 2e^{-(\Delta-2R)}.$$

Proof. Explicit swaps (w.p. π): ρS's margin on the pair $(c(s), c(\sigma s))$ is $-M_S(s)$, so the loss difference is $\ell(-M_S) - \ell(M_S) = M_S(s)$ *exactly*, by (2); expectation contributes $\pi\, \mathbb{E}[M_S]$. *Implicit collisions* (w.p. $\le \pi_{\text{impl}}$): the same identity applied to a within-class margin bounds the difference by R. *Unmatched*: both scorers' margins are $\ge \Delta - 2R$, so both losses are $\le e^{-(\Delta-2R)}$ and their difference at most twice that. □

Three measured phenomena from Part I become corollaries: the **linear dose–response** in π (the preference *is* linear, with slope $\mathbb{E}[M]$); the **λ–π equivalence** (Corollary K.3); and **cheap below-chance binding** — the exactly-reversed scorer sits only $\pi\, \mathbb{E}[M]$ away in loss, so a small perturbation (leakage, implicit bias) suffices to select it, consistent with the measured backwards-binding of learned MLP codes (§E).

K.3 Verification, and a loop closed with the frontier

Monte-Carlo verification: a factored toy world (M=6, K=4, all 2-scenes, N=512, images = code sum + Gaussian noise, 2×10^5 samples per point), scorers S^* (template), $\bar{S}^*$, ρS^*, sweeping $\pi \in [0, 0.5]$.

Figure 15 shows the result: exact linearity, and the matched-term constants recovered to three decimals once the unmatched offset is accounted for. The regime analysis closes an unexpected loop with the frontier. The offsets c_0, c_1 are large in the toy precisely because *template* scorers have large within-class spread R; the clean $\Delta - 2R$ regime of the theorems requires R small. But small within-class spread of real text towers is exactly what the frontier measurements of §H establish ($d \approx 0.24$ on the quartet probe): **the smoothness that caps binding (Theorem G.3) is the same measured property that certifies**

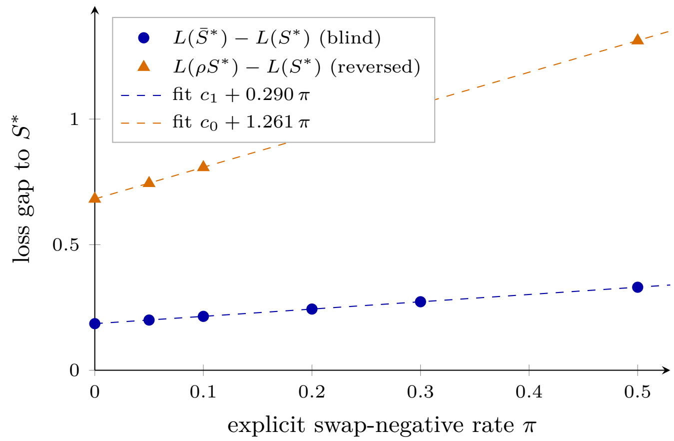


Figure 15: Verification of Theorems K.1–K.4 (noise 0.6). Both gaps are exactly linear in π. The toy sits in a low-separation regime, so unmatched contrasts contribute the π-independent offsets c_1, c_0; the decomposition $\text{gap}(\pi) = c + \pi(\text{matched difference} - c)$ recovers the theorems' matched constants precisely: $0.290 + c_1 = 0.476$ vs. predicted $\log 2 - \mathbb{E}[\ell(M^*)] = 0.477$, and $1.261 + c_0 = 1.943$ vs. predicted $\mathbb{E}[M^*] = 1.949$ (Monte-Carlo error ≈ 0.005). At noise 0.25: 0.5445 vs 0.5460 and 1.9455 vs 1.9511.

the throttle bounds are tight for deployed encoders. One measured quantity feeds two theorems.

K.4 Honest scope and open items

Not established: which solution training *selects* inside the sliver (a dynamics or implicit-bias question; the tempting symmetry route fails because the swap is not a vocabulary relabeling); the finite-sample version (an $\Omega(1/\pi^2)$ testing lower bound for detecting the correct binding sign is the natural target); the separation of instance binding from co-occurrence priors for non-exchangeable $\mathcal{D}$ (the "text prior" phenomenon); the multi-negative softmax version (the matched event becomes "swap present in batch").

L The depth ceiling of recursive composition

Part I's concept system is flat — a scene is a *set* of atoms, and every margin in §E is $\Theta(1/k)$ in the *number* of atoms pooled at one level. The homomorphism form (H) of §E.1 is recursive, and recursion is what makes language productive. This section reports what happens to binding under *nesting*: the margins obey an exactly geometric law in depth — proved below, and measured across three orders of magnitude — giving a logarithmic ceiling on the nesting depth a single pooled vector can resolve.

L.1 Recursive role-binding

Definition L.1 (Recursive role–filler composition)**.** Fix a branching factor b and position roles $r_1, \dots, r_b$ (random sign vectors). A depth-D structure is a full b-ary tree whose leaves carry filler codes (random unit sign vectors); each internal node's code is the normalized role-bound sum of its children,

$$\text{code}(v) \;=\; \frac{\sum_{j=1}^{b} r_j \odot \text{code}(\text{child}_j(v))}{\left\|\sum_j r_j \odot \text{code}(\text{child}_j(v))\right\|}.$$

This is the classical HRR/VSA recursion [10, 11] and the recursive instance of the pooled codes of §E (depth 1, $b = k$ recovers them). The *deep-edit margin* at depth D is $m(D) = 1 - \cos(\text{code}(T), \text{code}(T'))$ for T' a minimal edit of T (two leaves swapped, or one leaf replaced).

Two conventions for the roles will matter. In the *shared* scheme the b role vectors $r_1, \dots, r_b$ are drawn once and reused at every level — the reading of Definition L.1 above, and the classical HRR convention. In the *fresh* scheme a new independent role $r_{\ell,j}$ is drawn for each level ℓ and slot j. The exact law below holds for both, wherever no exact algebraic collision occurs; the finite-dimension theorem is stated for the fresh scheme, and §L.5 shows why: shared sign roles are self-inverse, and at certain depth/branching combinations they *alias* distinct structures exactly. Throughout we use two elementary facts about sign vectors $\rho \in \{\pm 1\}^N$, both immediate from $\rho_k^2 = 1$:

$$\text{(i)} \quad \langle \rho \odot x,\, \rho \odot y \rangle = \langle x,\, y \rangle, \qquad\qquad \text{(ii)} \quad \|\rho \odot x\| = \|x\|. \tag{3}$$

Sign-binding is an inner-product isometry: every distortion in what follows comes from *superposition* (adding b bound children) and *normalization*, never from the binding itself.

L.2 The contraction lemma and the exact law

The engine of the proof is one deterministic lemma: a single level of role-bound, normalized pooling maps the *slot deficits* of its children to their sum divided by b, up to an error controlled entirely by cross-slot inner products. No probability enters until §L.3; the lemma is exact bookkeeping.

Lemma L.2 (One-level contraction)**.** *Let $z_1, \dots, z_b$ and $z'_1, \dots, z'_b$ be unit vectors with $z_j = z'_j$ for $j \notin E$ (the* edited slots*); write $g_j := 1 - \langle z_j,\, z'_j \rangle$ for $j \in E$ and $G := \sum_{j \in E} g_j$ (throughout this section g_j, G denote slot deficits; the gain g of §E does not appear). Let $\rho_1, \dots, \rho_b$ be sign vectors, $P := \sum_j \rho_j \odot z_j$, $P' := \sum_j \rho_j \odot z'_j$, $y := P/\|P\|$, $y' := P'/\|P'\|$, and let the* slot crosstalk *be*

$$\varepsilon := \max_{i \neq j} \max\Big(\big|\langle \rho_i \odot z_i,\, \rho_j \odot z_j \rangle\big|,\ \big|\langle \rho_i \odot z_i,\, \rho_j \odot z'_j \rangle\big|,\ \big|\langle \rho_i \odot z'_i,\, \rho_j \odot z'_j \rangle\big| \Big).$$

If $(b-1)\varepsilon \le \frac{1}{2}$, then

$$\Big| \big(1 - \langle y,\, y' \rangle\big) - \tfrac{G}{b} \Big| \ \le\ 8(b-1)\varepsilon. \tag{4}$$

Proof. By (3)(i) the diagonal part of $\langle P,\, P' \rangle$ is $\sum_j \langle z_j,\, z'_j \rangle = b - G$; the $b(b-1)$ off-diagonal terms are each at most ε in magnitude, so $\langle P,\, P' \rangle = b - G \pm b(b-1)\varepsilon$. The same count gives $\|P\|^2 = b \pm b(b-1)\varepsilon$ and $\|P'\|^2 = b \pm b(b-1)\varepsilon$; hence, writing $u := (b-1)\varepsilon \le \frac{1}{2}$, $\|P\|\|P'\| = \sqrt{\|P\|^2 \|P'\|^2} = b(1 \pm u)$. Therefore

$$1 - \langle y,\, y' \rangle \ =\ \frac{\|P\|\|P'\| - \langle P,\, P' \rangle}{\|P\|\|P'\|} \ =\ \frac{G \pm 2bu}{b(1 \pm u)},$$

and since $G \le 2|E| \le 2b$,

$$\Big| \frac{G \pm 2bu}{b(1 \pm u)} - \frac{G}{b} \Big| \ \le\ \frac{u\,(G + 2b)}{b\,(1-u)} \ \le\ \frac{u \cdot 4b}{b/2} \ =\ 8u. \qquad \square$$

For the finite-dimension theorem we need a sharper form when the edits are *small* — the situation at every level above the edit, where the deficit has already been contracted. The refinement rests on two identities that are *exact*, with all approximation confined to a single square root.

Lemma L.3 (Refined contraction for small edits)**.** *In the setting of Lemma L.2, write $z'_j = z_j + \delta_j$ for $j \in E$ (so $\|\delta_j\| = \sqrt{2g_j}$), and set*

$$B := \sum_{j \in E} \sum_{i \neq j} \langle \rho_i \odot z_i,\, \rho_j \odot \delta_j \rangle, \qquad C_\delta := \sum_{\substack{i,j \in E \\ i \neq j}} \langle \rho_i \odot \delta_i,\, \rho_j \odot \delta_j \rangle, \qquad \bar{B} := B + \tfrac{1}{2} C_\delta.$$

Then, exactly*,*

$$\langle P,\, P' \rangle = \|P\|^2 - G + B, \qquad \|P'\|^2 = \|P\|^2 + 2\bar{B}, \tag{5}$$

and if $|\bar{B}| \le \|P\|^2/4$,

$$\Big| \big(1 - \langle y,\, y' \rangle\big) - \frac{G}{\|P\|^2} \Big| \ \le\ \frac{2G|\bar{B}|}{\|P\|^4} + \frac{|C_\delta|}{\|P\|^2} + \frac{3\bar{B}^2}{\|P\|^4}. \tag{6}$$

With the difference crosstalk $\tilde{\varepsilon} := \max\big|\langle \rho_j \odot \delta_j/\|\delta_j\|,\ \rho_i \odot w\rangle\big|$ *(maximum over $j \in E$, $i \neq j$, and $w \in \{z_i,\ \delta_i/\|\delta_i\|\}$, the last option when $i \in E$) one has $|B| \le (b-1)\tilde{\varepsilon}\sum_{j\in E}\sqrt{2g_j}$ and $|C_\delta| \le \tilde{\varepsilon}\big(\sum_{j\in E}\sqrt{2g_j}\big)^2 \le 2|E|\,G\,\tilde{\varepsilon}$: every error term carries at least one factor $\sqrt{g}$ — the injected error is proportional to the* size of the edit *— which is what makes the finite-N error law of Theorem L.8 multiplicative rather than additive.*

Proof. Let $\Delta := \sum_{j\in E}\rho_j \odot \delta_j$, so $P' = P + \Delta$. First identity: $\langle P,\ P'\rangle = \|P\|^2 + \langle P,\ \Delta\rangle$, and $\langle P,\ \Delta\rangle = \sum_{j\in E}\langle \rho_j \odot z_j,\ \rho_j \odot \delta_j\rangle + \sum_{j\in E}\sum_{i\neq j}\langle \rho_i \odot z_i,\ \rho_j \odot \delta_j\rangle = -G + B$, using (3)(i) and $\langle z_j,\ \delta_j\rangle = \langle z_j,\ z_j'\rangle - 1 = -g_j$. Second: $\|P'\|^2 = \|P\|^2 + 2\langle P,\ \Delta\rangle + \|\Delta\|^2$ with $\|\Delta\|^2 = \sum_{j\in E} 2g_j + C_\delta = 2G + C_\delta$, so $\|P'\|^2 = \|P\|^2 - 2G + 2B + 2G + C_\delta = \|P\|^2 + 2\bar{B}$.

For (6), set $s^2 := \|P\|^2$ and $x := 2\bar{B}/s^2$, so $|x| \le \frac{1}{2}$. Since $\sqrt{1+x} = 1 + \frac{x}{2} - \xi$ with $0 \le \xi \le 0.36\,x^2$ on $|x| \le \frac{1}{2}$ (Taylor with the second derivative of $\sqrt{1+x}$ maximized at $x = -\frac{1}{2}$),

$$\|P\|\|P'\| - \langle P,\ P'\rangle = s^2\sqrt{1+x} - \big(s^2 - G + B\big) = \bar{B} - s^2\xi + G - B = G + \tfrac{1}{2}C_\delta - s^2\xi,$$

and $\|P\|\|P'\| = s^2\sqrt{1+x}$, so

$$\Big|\big(1 - \langle y,\ y'\rangle\big) - \frac{G}{s^2}\Big| \ \le\ \frac{G}{s^2}\Big|\frac{1}{\sqrt{1+x}} - 1\Big| + \frac{|C_\delta|/2 + s^2\xi}{s^2\sqrt{1+x}} \ \le\ \frac{G}{s^2}\,|x| + \frac{3}{2}\Big(\frac{|C_\delta|}{2s^2} + 0.36\,x^2\Big),$$

using $\big|1/\sqrt{1+x} - 1\big| \le |x|$ and $1/\sqrt{1+x} \le \frac{3}{2}$ on $|x| \le \frac{1}{2}$. The three resulting terms are $2G|\bar{B}|/s^4$, at most $|C_\delta|/s^2$, and $0.54\,(2\bar{B}/s^2)^2 \le 3\bar{B}^2/s^4$. ☐

Theorem L.4 (The exact depth law)**.** *In the zero-crosstalk idealization ($\varepsilon = \tilde{\varepsilon} = 0$ at every level — the $N \to \infty$ limit of the random model), for the depth-D full b-ary code under either role scheme, provided all bound products appearing in the construction are distinct (see §L.5 for the shared-role exceptions):*

$$m_{\text{swap}}(D) \ =\ 2 \cdot b^{-D}, \qquad m_{\text{replace}}(D) \ =\ 1 \cdot b^{-D}, \tag{7}$$

for both the sibling swap (two leaves exchanged under one deepest parent) and the far swap (two leaves exchanged whose lowest common ancestor is the root). More generally, an edit whose slot deficits at level ℓ_0 sum to G has root margin $G \cdot b^{-(D-\ell_0)}$: the margin depends only on the total deficit and the number of contraction levels, not on where in the tree the edit sits.

Proof. With $\varepsilon = 0$, Lemma L.2 is exact: one level maps slot deficits summing to G to a parent deficit of exactly G/b; induct upward.

Sibling swap. The exchanged leaves u, v occupy slots $1, 2$ of one level-1 parent: apply Lemma L.2 with $E = \{1, 2\}$, $z_1 = u$, $z_1' = v$, $z_2 = v$, $z_2' = u$; distinct idealized primitives are orthogonal, so $g_1 = g_2 = 1$, $G = 2$, and the parent deficit is $2/b$. Each of the $D-1$ levels above contracts a single slot's deficit by exactly $1/b$: $m = (2/b) \cdot b^{-(D-1)} = 2b^{-D}$.

Far swap. Each of the two subtrees hanging off the root sees its leaf *replaced* by a vector orthogonal to it: base deficit 1, contracted through $D-1$ levels of its subtree to $b^{-(D-1)}$ at the root's child. At the root, $E = \{\text{left}, \text{right}\}$ and $G = 2b^{-(D-1)}$, giving $m = G/b = 2b^{-D}$.

Replace. One leaf replaced: base deficit 1, contracted D times: b^{-D}. The general statement is the same induction; location independence is immediate since the lemma's conclusion depends only on G and b. ☐

Remark L.5 (Why the constant is exact, and why sibling equals far). The G/b term suffers no b-dependent loss because sign-binding preserves each slot's inner product exactly, by (3)(i), while normalization divides by $\|P\|\|P'\| = b$ exactly at $\varepsilon = 0$: the law is the ratio (per-slot edit deficit)/(slots superposed), iterated. Depth 1 with $b = k$ recovers the depth-1 constants — swap $2/k$ (Theorem E.7) and replace $1/k$ (its companion computation): the depth law is that analysis composed with itself. And the equality of sibling and far swaps — which surprised us as a measurement — is transparent here: a far swap is two *replacements* in parallel branches, and $1+1$ contracted through D levels equals 2 contracted through D levels, by location independence.

L.3 Finite dimension: a multiplicative error law

Finite N replaces exact orthogonality by crosstalk of scale $\sqrt{1/N}$. The honest statement is for the fresh role scheme; $Q \le b^D + 1$ counts the distinct primitives (the b^D leaves plus the replacement filler).

Lemma L.6 (Level-0 crosstalk)**.** *With probability at least $1-\eta$, all pairwise inner products among the Q distinct primitives are at most $L_0 = \sqrt{2\log(2Q^2/\eta)/N}$ in absolute value, and by (3)(i) the bound survives binding by any fixed roles. (Hoeffding plus a union bound over at most Q^2 pairs, exactly the argument of Lemma E.5.)*

Lemma L.7 (Crosstalk propagation)**.** *Condition on all codes at level $\ell-1$ and on the event that every such code z — and every normalized difference $\delta/\|\delta\|$ of codes along the edited paths — satisfies $\|z\|_4^4 \le C_4/N$, where $\|\cdot\|_4$ is the coordinatewise 4-norm (the ℓ_4 hypothesis). Under the fresh scheme, for distinct slots $i \ne j$ of one parent the level-ℓ roles give $\langle \rho_i \odot z_i,\, \rho_j \odot z_j\rangle = \sum_k \sigma_k\,(z_i \odot z_j)_k$ with $\sigma := \rho_i \odot \rho_j$ a uniform sign vector independent of the codes, so by Hoeffding, with probability $1-\eta'$,*

$$\left|\langle \rho_i \odot z_i,\, \rho_j \odot z_j\rangle\right| \;\le\; \|z_i \odot z_j\|\,\sqrt{2\log(2/\eta')} \;\le\; \|z_i\|_4\,\|z_j\|_4\,\sqrt{2\log(2/\eta')} \;\le\; \sqrt{\frac{2C_4\log(2/\eta')}{N}},$$

and on the same event the same argument bounds the difference crosstalk $\tilde\varepsilon$ (coefficient vectors $z_i \odot (\delta_j/\|\delta_j\|)$, and likewise for the δ-against-δ pairs). (Proved, on the stated event.) Propagation of the ℓ_4 hypothesis: *at level 0, $\|z\|_4^4 = 1/N$ exactly, and a normalized primitive difference $(u-v)/\|u-v\|$ has $\ell_4^4 = (2+o(1))/N$ directly. For the inductive step, sibling codes are functions of disjoint leaf sets, hence independent conditional on the roles, with sign-symmetric entries; a direct fourth-moment computation for independent mean-zero slot entries gives*

$$\mathbb{E}\,\|y\|_4^4 \;\le\; \frac{3 + C_4/b}{N}\,\big(1+o(1)\big),$$

whose fixed point $C_4(b) = \frac{3b}{b-1}\,(1+o(1))$ is bounded uniformly in D: the fourth moment does not grow with depth. (The expectation step is proved; the high-probability form of this inductive step — concentration of $\|y\|_4^4$, of its analogue for the normalized differences along the edit path (which obey their own recursion through (5)), and of $\|P\|$ around their conditional means, e.g. by McDiarmid over the bN fresh role signs — is Outlined: standard but bookkeeping-heavy, and it is the single unproved step in this section.)

Theorem L.8 (Finite-N depth law, fresh roles)**.** *There are absolute constants C, C' such that for the fresh scheme with $D \ge 2$ and $N \ge C\,b^2D^2\log(bQ/\eta)$, with probability at least $1-\eta$, simultaneously for every edit of Theorem L.4 (depth 1 is the finite-N regime of Theorem E.7 itself),*

$$m(D) \;=\; m_{\text{ideal}}(D)\cdot(1 \pm \kappa_N), \qquad \kappa_N \;\le\; C'\sqrt{\frac{b^D\,\log(bQD/\eta)}{N}},$$

where m_{ideal} is the value in (7). In particular the products $m \cdot b^D$ are constant to relative error $o(1)$ whenever $b^D \ll N/$polylog.

Proof (given the Outlined step of Lemma L.7). Work on the intersection of the events of Lemmas L.6 and L.7, union-bounded over the $O(b^2D)$ slot pairs adjacent to the edit paths of the three canonical edit types and over the levels; on it, every crosstalk appearing below is at most $\varepsilon_N := \sqrt{2C_4\log(Cb^2D/\eta)/N}$ (and $\tilde\varepsilon \le \varepsilon_N$), and every base inner product is at most L_0. Run the induction of Theorem L.4 carrying errors, applying Lemma L.3 at *every* level including the base: its identities (5) are exact for edits of any size, its smallness hypothesis $|\bar B| \le \|P\|^2/4$ holds on the event (each ingredient of $\bar B$ is $O(b\varepsilon_N)$ against $\|P\|^2 \ge b/2$), and at every level $|E| \le 2$ with $\sum_{j\in E}\sqrt{2g_j} \le 2\sqrt{3}$. The base deficit itself is $G_{\text{ideal}}(1 \pm L_0)$. Dividing (6) by the level's ideal deficit and using $s^2 = b(1 \pm (b-1)\varepsilon_N)$, $|B| \le (b-1)\tilde\varepsilon\sum_{j\in E}\sqrt{2g_j}$, and $|C_\delta| \le 2|E|G\tilde\varepsilon$, each level contributes a *relative* error

$$e_\ell \;\le\; C_0\Big[(b-1)\,\varepsilon_N \;+\; \frac{b-1}{b}\,\tilde\varepsilon\sum_{j\in E}\sqrt{2g_j} \;+\; \tilde\varepsilon \;+\; (b-1)^2\tilde\varepsilon^{\,2}\Big]$$

for an absolute constant C_0 — the four terms coming from the normalization crosstalk, the $2G|\bar{B}|/s^4$ term, the $|C_\delta|/s^2$ term measured relative to G/s^2, and the $3\bar{B}^2/s^4$ term. Since the deficits decay geometrically along the path, $\sum_\ell \sum_{j\in E}\sqrt{2g_j}$ is bounded by an absolute constant, so summing over the at most $2D$ contraction steps of the two edit paths, $\sum_\ell e_\ell \le C_1(1+bD)\,\varepsilon_N \le \frac{1}{2}$, the last inequality by $N \ge Cb^2D^2\log(bQ/\eta)$ with C large; then $\prod_\ell(1+e_\ell)-1 \le 2\sum_\ell e_\ell$, and the base factor contributes a further L_0. Finally $(1+bD) \le 3\,b^{D/2}$ for all $b \ge 2$, $D \ge 2$, and $L_0 \le \sqrt{2}\cdot\sqrt{\log(bQD/\eta)/N}$, so the accumulated relative error is at most $C'\sqrt{b^D\log(bQD/\eta)/N}$. □

Remark L.9 (The proof gives more than the statement). The accumulation above in fact bounds the relative error by an absolute constant times $(1+bD)\,\varepsilon_N + L_0$ — polynomial in D, not $b^{D/2}$ — because Lemma L.3 makes each level's injected error proportional to $\sqrt{g_\ell}$ and the deficits decay geometrically. We state the theorem in the more conservative form; both forms rest on the same single Outlined step.

Remark L.10 (The error is multiplicative — measured). Theorem L.8 predicts the *relative* spread of m to be roughly flat in D at fixed N, with the *absolute* spread shrinking like b^{-D}; an additive error law would instead show flat absolute spread. Measured (per-seed relative standard deviation of m_{sib}, $b=2$, $N=4096$, 24 seeds): 2.6% at $D=1$ against 3.5% at $D=10$, never above 6% at any intermediate depth, while the absolute standard deviation shrinks by a factor of ≈370; for reference, $1/\sqrt{N}=1.6\%$. The side-by-side sweep reported in §L.5 shows the same flatness (0.8%–2.4% across all 32 of its configurations, both role schemes). This is a structural confirmation of the contraction mechanism, not merely of its conclusion.

L.4 The law, measured

Simulation, shared roles ($b=2$: $D\le 10$; $b=3$: $D \le 7$; $N \in \{256,\dots,65{,}536\}$; 24 seeds per configuration; three edit types): the products $m\cdot b^D$ sit at the predicted constants 2 (both swap types) and 1 (replace), within 0.014 and 0.006 respectively at $N = 65{,}536$ over every configuration (the aliased shared-role far-swap cells of §L.5 excepted) — exactly the depth-1 constants of Theorem E.7 and its replace companion, composed multiplicatively per level, as Theorem L.4 requires. At smaller N the deviations grow — maximum $|m_{\text{sib}}\,b^D-2|$ of 0.20, 0.10, 0.034, 0.014, 0.012 across $N=256, 1024, 4096, 16384, 65{,}536$, the other edit types behaving analogously at their own scale — and these are the finite-N corrections whose size Theorem L.8 bounds and whose multiplicative signature Remark L.10 confirms. A follow-up sweep under the *fresh* scheme ($N=16{,}384$, both schemes side by side, 24 seeds, 32 configurations in all) lands every swap product in $[1.985, 2.014]$ and every replace product in $[0.993, 1.005]$, including as swaps the shared-role aliased cases that the fresh scheme repairs (§L.5).

Crossing the law with a detection floor yields the ceiling:

Corollary L.11 (Resolvable depth)**.** *Fix a readout whose decision floor is* $\nu(N) = c\sqrt{\log(1/\eta)/N}$ *— the primitive crosstalk scale of Lemma L.6, the floor of any linear readout against a stored template. On the event of Theorem L.8 (depth 1 being covered exactly by Theorem E.7), binding at depth D is resolvable if* $2b^{-D}(1-\kappa_N) > \nu(N)$ *and unresolvable if* $2b^{-D}(1+\kappa_N) < \nu(N)$*; either threshold reads*

$$D \;\le\; D^*(N) \;=\; \frac{\log N}{2\log b} + O(1), \tag{8}$$

a staircase gaining one level per b^2*-fold increase of* N*.* (Proved given Theorem L.8; the staircase itself is a measurement.)

Verified exactly: with floor $10/\sqrt{N}$, $D^* = 1,2,3,4,5$ at $N=256,\dots,65{,}536$ for $b=2$ (one level per 4× dimension) and $1,1,2,2,3$ for $b=3$ (Figure 2). At CLIP's $N=512$ and linguistic branching: the pure crossing gives 4.5 ($b=2$) and 2.8 ($b=3$), the measured $10/\sqrt{N}$ floor $D^*=2$ and 1 — single digits either way: **the ceiling sits at ordinary natural-language nesting depth**, which is precisely the regime where companion grammar-vs-flat experiments (depth extrapolation, CLEVR-CoGenT, tree-structured composers; reported separately) locate the advantage of structured models.

L.5 Structural aliasing: some deep edits are invisible at any dimension

The simulation also revealed a phenomenon that goes beyond attenuation.

Proposition L.12 (Aliasing for self-inverse roles)**.** *With sign-vector roles shared across levels, roles are self-inverse under $\odot$ ($r \odot r = \mathbf{1}$), so a leaf's coefficient in the (pre-normalization) root expansion depends only on the* parity profile *of its role chain. Two leaves whose chains have equal parity profiles have identical coefficient vectors; exchanging them leaves the unnormalized root code (equivalently, any code normalized by deterministic per-level scalars)* exactly *unchanged — margin $\equiv 0$ at* any *dimension, with no orthogonality used. Under per-node normalization the exchange survives only in the norm-mismatch scalars, so the residual margin is crosstalk-sized, carrying no $2b^{-D}$ term.*

Proof. Unrolling the recursion, each leaf x enters the root as $(\bigodot_{\text{path}} r_j) \odot x$ times a positive scalar — the product of the inverse norms along its path, level-uniform under deterministic normalization and node-dependent (with crosstalk-sized deviations) under per-node normalization. Since $r \odot r = \mathbf{1}$, the path product depends only on which roles appear an odd number of times. Equal parity profiles $\Rightarrow$ equal coefficient vectors $\Rightarrow$ the two leaves' contributions commute with exchange — exactly in the deterministic case, and up to the norm mismatch in general. □

The simulation confirms it: for $b = 3$ at even depths, the far swap (both leaves at position-1 chains of equal parity) has margin two to five orders of magnitude below the law's prediction at every N measured, shrinking with N as pure crosstalk must. Flat sign-role binding does not merely attenuate deep structure — it *aliases* some of it. Fresh per-level roles repair it, measured directly: in the side-by-side sweep at $N = 16{,}384$ (24 seeds), the aliased configurations ($b = 3$; $D = 2, 4, 6$) have $m_{\text{far}} \cdot b^D$ of 0.0001, 0.0003, 0.0006 under shared roles — and 1.994, 2.000, 2.000 under fresh roles, exactly the generic law (7). Theorem L.8 is stated for precisely that scheme; aliasing stands as a separate warning for self-inverse shared-role binding.

L.6 Status and positioning

The proof route this section originally recorded only as an outline — induction on the Theorem E.7 computation, with Lemma E.5-style crosstalk control — is now carried out in §L.2–§L.3. The status is as follows: Lemmas L.2–L.3 and Theorem L.4 are proved in full; Theorem L.8 is proved modulo the single outlined concentration step inside Lemma L.7; Corollary L.11 is proved given Theorem L.8; the law's empirical confirmation, the staircase, the multiplicative-error signature, and the fresh-role repair of aliasing are measured. The sequence-memory analysis of Frady, Kleyko and Sommer [12] supplies closely related crosstalk accounting, with their contraction λ^K (from recurrent dynamics) replaced here by b^{-D} (from per-node normalization); their theory has no per-level margin law and no two-structure discrimination — the gap Theorems L.4–L.8 close. Plate's classical account [10] describes qualitative degradation but draws the opposite conclusion — that deep structure remains workable, and that chunking removes the limit — chunking being exactly the structured-scorer escape branch. *Scope caution:* as a no-go the statement is class-relative (bounded-norm homomorphic combinators); an unconstrained encoder can look up any finite set of deep captions, so the general-encoder version requires a margin-complexity argument or a measurable hypothesis in the style of Theorem G.3.

M What does *not* limit binding: a conjecture refuted

Pinning down the reason for a failure requires eliminating candidate reasons. This section records a plausible, literature-adjacent conjecture that we formulated and then *refuted*; the refutation sharpened the map.

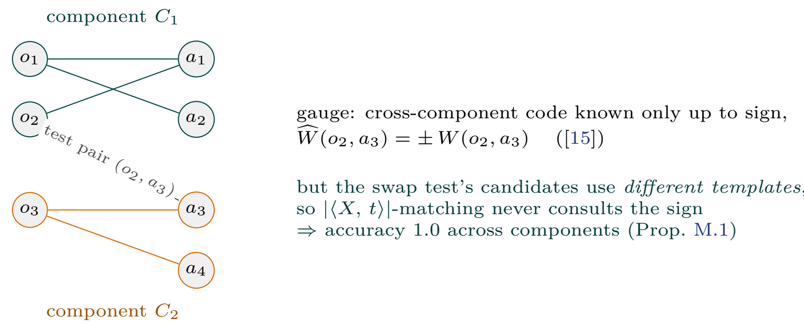


Figure 16: The refuted conjecture. Across disconnected exposure components the factored code of an unseen pair is identified only up to sign — but the swap decision compares templates with different *supports*, and a sign-agnostic matcher decides it without ever resolving the gauge. Coverage (both atoms seen), not connectivity, is the true information threshold.

M.1 The conjecture

Training exposes a set $P \subseteq O \times A$ of bound pairs — the *exposure graph* G, bipartite on objects and attributes. In a factored world ($W(o,a) = u(o) \odot v(a)$), classical rank-one completion theory [15] says the codes are identified only up to a per-connected-component gauge, and cross-component products only up to a sign. The conjecture: *binding generalization to a pair whose endpoints lie in different components of G is information-theoretically impossible — no learner can beat chance — so compositional generalization undergoes a phase transition at the connectivity threshold of G.* (Adjacent published work: a connectivity-type compositional-support condition is assumed for identification by Schug et al. [14]; a percolation model, established by analogy rather than learner-side theorems, is given by Lubana et al. [13].)

M.2 The refutation

Proposition M.1 (The sign-agnostic matcher succeeds across components)**.** *In the noiseless factored world, let both atoms of the unseen test pairs be exposed (possibly in different components of G), and let $\widehat{u}, \widehat{v}$ be recovered by gauge propagation (exact up to a sign per component). For the swap test on the recombined scene $s = \{(o_1, a_1'), (o_2, a_2')\}$, score each candidate caption by* $\mathrm{score}(T) = |\langle X, t_1\rangle| + |\langle X, t_2\rangle|$ *where X is the image code and t_1, t_2 the candidate's pair templates. On the primitive-control event of Lemma E.5, the correct caption's score is $\geq 2 - O(kL^*)$ while the swapped caption's is $O(kL^*)$: the decision is correct regardless of the unresolved signs, because the two candidates' templates have different supports and the modulus discards the gauge.*

Proof. $X = W(o_1, a_1') + W(o_2, a_2')$. The correct candidate's templates are $\pm W(o_1, a_1'), \pm W(o_2, a_2')$ (signs unknown): $|\langle X, t_i\rangle| = |1 + O(L^*)|$ each. The swapped candidate's templates are $\pm W(o_1, a_2'), \pm W(o_2, a_1')$ — distinct atom pairs, so every inner product with X is a primitive-sized quantity $O(L^*)$ (Lemma E.5). □

Simulation ($M{=}K{=}40$, $N{=}2048$, gauge-propagation learner, exposure density swept through the fragmented regime): sign-agnostic accuracy $= 1.000$ at every density, *including when 0% of test pairs share a component.* The conjecture is dead for swap tests: **coverage (both atoms seen), not connectivity, is the information threshold** — precisely the occupancy law that Proposition F.2(ii) reports as measured.

M.3 What survives

1. **A below-chance signature of gauge commitment.** The sign-*naive* learner (one that commits to its arbitrary per-component gauge) scores 0.27–0.37 — systematically *below* chance — on cross-component tests. An estimator that commits to

an unidentified convention binds *backwards*, echoing both the measured backwards-binding of leakage-prone codes (§E) and the selection-inside-the- sliver picture of §K; cf. the min-degree implicit-bias analysis of Abbe et al. [16].

2. **The occupancy law as a theorem-target.** The measured curve $\mathrm{acc}(\rho) \approx \frac{1}{2} + \frac{1}{2}(1 - e^{-\rho K})(1 - e^{-\rho M})$ is an isolated-atom (coupon-collector) asymptotic, derivable rather than fitted — and distinct from the giant-component threshold of the percolation narrative [13], a reconciliation worth writing out.
3. **A design note.** A genuine connectivity no-go would require a task whose answer *depends on the gauge* (sign-sensitive retrieval with sign-flipped distractors); benchmark-style swap tests are not such a task.

N The updated map, and open problems

N.1 The causal story, as the results now support it

Binding is representable and cheap (factored codes, $N = O(k^2 \log MK)$, §E); *the geometry permits it* (Corollary C.4); *the population objective, at universal capacity, even prefers it* (Lemma F.1). Real systems fail because:

1. **the objective's reward for binding is a sliver** of width $(\pi + \pi_{\mathrm{impl}}) \log 2$ nats — now proved (Theorems K.1–K.4) — so training pressures of any size beyond that (capacity, regularization, early stopping, implicit bias) decide the outcome; and
2. **learned composition codes leak** (measured $\ell \approx +0.17$; §E), and inside the sliver even the exactly-reversed solution is $\pi\,\mathbb{E}[M]$-cheap — which is why below-chance binding is observed.

Behind these *contingent* causes stand *terminal* limits that survive all fixes: the smoothness frontier (Theorem G.3; measured slack today) and the depth ceiling ($D^* = \Theta(\log N/\log b)$, §L; sitting at natural-language depth for CLIP-sized models). And two candidate causes are *eliminated*: dimension (§C) and exposure-graph connectivity (§M).

N.2 Open problems (Part II update)

1. **Selection inside the sliver.** Which solution do SGD dynamics pick when the objective is indifferent at the $(\pi + \pi_{\mathrm{impl}})$ scale? The below-chance signatures (leakage; gauge commitment) suggest the tie-breakers are not benign.
2. **Finite-sample throttle.** An $\Omega(1/\pi^2)$ lower bound for detecting the correct binding sign from rate-π swap-contrast events (two-point testing).
3. **The depth-law concentration step.** The law (7) is now Proved (Theorems L.4–L.8) except for one concentration estimate — the high-probability inductive step of Lemma L.7 — which remains Outlined; close it. Then the margin-vs-depth measurement on *real* nested captions through real text towers — the depth analogue of the §H zoo sweep.
4. **Instance binding vs. co-occurrence priors.** For non-exchangeable $\mathcal{D}$, separate what $\pi = 0$ training can learn (the prior $p(a \mid o)$ — the benchmark "text prior") from what it cannot (instance pairing).
5. **Part I items that remain open:** the anchor-robustness appendix for the frontier; naturalistic multi-relation probes; the NegCLIP three-axis confound.

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